HS 3410 · END-SEMESTER EXAM PREP

Modern Science in India — Simple Notes

Easy-language notes on what the British and Indians wanted from science and technology, why they wanted it, why they sometimes didn't, and what the author is really trying to tell us.

Based on: David Arnold, *Science, Technology and Medicine in Colonial India* (The New Cambridge History of India III.5, Cambridge University Press, 2000).
Chapters covered: 1. Introduction: Science, Colonialism and Modernity · 2. Science under the Company · 4. Technologies of the Steam Age · 5. Imperial Science and the Indian Scientific Community.

Ch 1 · Science, Colonialism & Modernity
Ch 2 · Science under the Company
Ch 4 · Technologies of the Steam Age
Ch 5 · Imperial Science & Indian Scientists
Essay plans + Quick revision

HOW TO USE THESE NOTES (2 DAYS BEFORE THE EXAM)

1. **Read "The Big Picture" first.** If you only remember those pages, you can still write a decent essay.
2. **Then each chapter.** Every chapter follows the same pattern: the story in simple words → why the British wanted it / why they held back → the Indian side → **Arnold's main point** → examples to remember.
3. **Then the Essay Plans.** All 6 past test questions are planned out, plus 10 likely new ones. Practise writing at least 2 full answers.
4. **Night before:** only read the Quick Revision sheets at the end (dates, people, historians, quotes, numbers).

Page numbers like **(p. 25)** point to the page in Arnold's book, so you can check a quote quickly if you need to.

Contents

The Big Picture: the whole book in 2 pages 3

Introduction: Science, Colonialism and Modernity 6

1.1 Where Arnold starts 6

1.2 India's scientific traditions 6

1.3 "Colonial science": Basalla and his critics 8

1.4 Science as modernity 9

1.5 Chapter 1 in a nutshell 10

Science under the Company 11

2.1 The Company and modern science grew up together 11

2.2 The Company as patron: control, stinginess, self-interest 11

2.3 The great surveys: Mackenzie and Buchanan 12

2.4 Mostly a spare-time, "gentlemanly" science 13

2.5 How colonial science was organised 13

2.6 Science and Orientalism: from respect to rejection 14

2.7 The Great Trigonometrical Survey (GTS) 15

2.8 Geology and zoology 16

2.9 Botany: profit + philanthropy 16

2.10 Science in the tropics: India as "tropical" 17

2.11 Chapter 2 in a nutshell 17

Technologies of the Steam Age 19

4.1 Arnold's opening argument 19

4.2 Textiles: decline, adaptation and survival 19

4.3 Mines and metallurgy 20

4.4 Shipbuilding and steamships 21

4.5 Roads and railways 22

4.6 Telegraphs 23

4.7 Irrigation 23

4.8 Technology, ideology and resistance 24

4.9 Steam, Swadeshi and Swaraj: is industry right for India? 25

4.10 Chapter 4 in a nutshell 25

Imperial Science and the Indian Scientific Community 27

5.1 The big shift after 1858 27

5.2 Science under the bureaucrats (the ICS) 27

5.3 "Imperial science" under Elgin and Curzon 28

5.4 Medical science: from neglect to "imperial" research 29

5.5 Forestry and agriculture 30

5.6 The rise of an Indian scientific community 31

5.7 Chemistry, physics and plant physiology: the Indian breakthrough 33

5.8 Chapter 5 in a nutshell 34

Essay Plans: Past Questions + Likely Questions 36

A. The 6 past test questions 36

B. Likely new questions (shorter plans) 43

Quick Revision Sheets 48

R1. Timeline 48

R2. Who's who — people to name in essays 50

R3. Historians and their arguments (name-drop these!) 51

R4. Quotes bank (short and powerful) 52

R5. Numbers that impress examiners 54

START HERE

The Big Picture: the whole book in 2 pages

Arnold's whole argument in one line: science in colonial India was never just "Western knowledge spreading to India". It was a **social and political** thing, used by the British for **profit, control and pride**, held back when it threatened British interests, shaped by **Indian knowledge, labour and conditions**, and later taken up by Indians to build their **own modernity and nation**.

Arnold's 6 big messages (use these in every essay)

1. **Science is not neutral.** It is "a highly social activity" (p. 1). It carries money, power, politics and ideas about who is "superior". So we must always ask: *who* is doing science, *for whom*, and *why*?
2. **It was not a one-way gift.** The old story says the West "diffused" (spread) science into a passive India. Arnold says no — it was an **interaction / dialogue**. Indians gave knowledge, skills, labour and materials; the British had to change their methods to fit India.
3. **But the interaction was unequal.** It happened inside colonial rule. Britain and its scientists stayed on top; Indians were mostly kept in lower posts. Arnold calls this the "political character of science".
4. **For the British, science was mostly a tool and a symbol.** A tool for revenue, trade, war and control. A symbol of being "civilised" and superior (the *civilising mission*).
5. **"Denial was often as important as diffusion"** (p. 92). The British also *stopped* Indians from getting science and technology when it could harm British interests (shipbuilding, metals, locomotives, top science jobs).
6. **Indians were not passive.** Some embraced science as the key to progress and nationhood (Mahendralal Sircar, P. C. Ray, J. C. Bose, C. V. Raman, Nehru). Some mixed it with Indian ideas. Some rejected industrial modernity altogether (Gandhi).

How the story moves (3 stages)

STAGE	WHAT SCIENCE / TECHNOLOGY MEANT	CHAPTER
Company period c. 1757–1858	Science as **"hobby + tool"**. Surveys, botany, geology, astronomy, mapping. Done mostly by **army engineers and surgeons** in their spare time. The Company cared only when there was **profit, war or prestige** in it — a "fitful flirtation".	Ch 2
Steam age c. 1820s–1900s	Technology as **power and proof of superiority**: steamships, railways, telegraph, canals, roads, mints. Shaped by Indian conditions and labour. Benefits went mostly to Britain. Debates on whether India should industrialise.	Ch 4
Crown / "imperial science" 1858–1920s (peak under Curzon, 1899–1905)	Science tightly tied to the **state**: census, gazetteers, scientific services, research institutes. But controlled by bureaucrats, racially exclusive and mostly "applied". **At the same time**, Indians build their **own scientific community** (IACS 1876, Ray, Bose, Raman, Indian Science Congress 1914).	Ch 5
Throughout	The big question: **whose science, whose modernity?** Basalla's diffusion model vs Arnold's interactive, power-aware model. British "civilising mission" vs Indian nationalism vs Gandhi.	Ch 1

The "motive" question in one table

WHY THE BRITISH WANTED SCIENCE & TECHNOLOGY	WHY THE BRITISH HELD IT BACK / DIDN'T CARE
• **Money** — revenue, trade, cash crops (indigo, cotton, tea), coal, timber. • **War & security** — maps, roads, railways, telegraph to move troops and get news fast. • **Knowing India to rule India** — surveys, census, gazetteers ("governmentality"). • **Prestige & legitimacy** — "we are civilised, we bring progress"; very important after 1857. • **Famine** — botanic gardens, railways, canals, agricultural research were partly answers to famine. • **Personal interest** of officers, surgeons, governors-general (Bentinck, Dalhousie, Curzon).	• **Cost** — the Company was "cash-conscious"; science was secondary to revenue, law and order. • **Competition** — didn't want India to compete with British copper, ships, locomotives, machinery. • **Security** — metal-working and arms could help Indian rebels (Arms Act 1878, mines closed). • **Racism** — Indians seen as "unfit" for original science. • **Romantic idea** — India should stay a land of peasants, princes and artisans. • **Bureaucrats over scientists** — the ICS kept scientists under control.
WHY INDIANS WANTED SCIENCE	**WHY SOME INDIANS DOUBTED IT**
• To understand the source of Western power and progress (Sircar). • To reform society and religion by reason (Western-educated elite). • National pride: prove Indians can do "pure" science (Bose, Ray, Raman). • Self-reliance and industry (Swadeshi, Tata, Bengal Chemicals).	• Modernity looked like copying the West — always "second-best". • Too materialistic (Keshab Chandra Sen). • Machines and railways bring poverty, disease, slavery (Gandhi, *Hind Swaraj*, 1909). • Intrusive colonial measures (plague control) caused resistance.

Words you must know (in simple English)

WORD	SIMPLE MEANING
Metropole / metropolitan / "centre"	Britain (London) — the home of the empire: Royal Society, Kew Gardens, Court of Directors, British universities.
Periphery	The colony — India. Seen as the "edge" of the scientific world.
Diffusion / diffusionist	The idea that science simply "spreads" from the West to the rest, like ink in water. Arnold criticises this.
Colonial science	Science done in a colony. For Arnold: science (by Europeans *and* Indians) that is always tied up with colonial power.
Orientalism / Orientalists	European scholars who studied Asian languages, texts, history (e.g. Sir William Jones). They admired ancient India but saw modern India as declined.
Utilitarian	Done for practical use / profit, not just for knowledge.
"Improvement"	18th–19th c. British idea: making land, crops and people more productive through knowledge.
Governmentality (Foucault)	When the state collects detailed information on land, people and resources so it can manage and control them.

Panopticon (Foucault, used by Edney)	A system where the ruler can "see everything". Used for the idea that maps/surveys let the British "see and know" all of India.
Hegemony	Domination not only by force but through ideas that people come to accept.
Civilising mission	British claim that they ruled India to "civilise" and improve it. Science and technology were their main proof.
Tropicality	The idea that tropical lands (like India) are different from Europe — rich nature but disease, famine and "weak" people who need European help.
Indigenous knowledge	Local Indian knowledge — of plants, medicine, wells, ships, metals, forests, farming.
Bhadralok	The educated Bengali middle class (mostly Brahmin, Kayastha, Vaidya castes).
ICS / IMS	Indian Civil Service (top administrators) / Indian Medical Service (army & government doctors).
Swadeshi / Swaraj	"Own-country made" (the 1905 movement to make Indian goods) / self-rule.
Anicut	A dam/weir across a river (South Indian word), e.g. the Grand Anicut on the Cauvery.
Laissez-faire	Government should not interfere in the market. (So in famine, the British built railways instead of controlling grain prices.)

CHAPTER 1

Introduction: Science, Colonialism and Modernity

Book pages 1–18 · This is the "theory" chapter — use it in the intro and conclusion of almost any essay.

In one line: Science in colonial India was not just "Western science arriving in India". It was shaped by three things together — **India's own traditions**, **colonial power**, and the **fight over what "modern" means**.

1.1 Where Arnold starts

- Science is **"a highly social activity"**. It cannot be "sealed off" from the values of the society where it is done. It looks different in different times and cultures — it is "contingent and culturally specific".
- So, a history of science in India must also be **a history of India** — not just a story of the West throwing its science onto India.
- Arnold says three things shaped science, technology and medicine in India over these 200 years (late 1700s to 1947):
  1. **India's own traditions** of science, technology and medicine, and what they left behind for the British period.
  2. **Western ("colonial") science** as practised in India — and its *double* relationship: to the colonial government in India, and to "metropolitan" science in Europe.
  3. **Science as modernity** — science became the main sign of being "modern", and both British (imperial ideology) and Indians (nationalism) fought over it.

The chapter discusses these three one by one. So will we.

1.2 India's scientific traditions

(a) There was no single "Indian science"

India had many traditions — oral and written, Hindu and Muslim, regional, and many borrowed from outside. Even inside the "Hindu" tradition there were many strands. For example, **Vedic astronomy** was linked to religion (you needed to know solstices and equinoxes to time the sacrifices correctly). Later, astronomy, trigonometry and algebra moved partly away from religion. The ***Surya Siddhanta*** (around AD 400) discussed planets, eclipses and instruments like the armillary sphere — but also cosmology and "malignant aspects of the sun and moon". So **observation and belief lived side by side**.

(b) The old "Orientalist" story: golden age → decline → British rescue

European Orientalist scholars (late 1700s – early 1800s) told a three-part story. Arnold calls it the **Orientalist "triptych"** (a picture in three panels):

1. **Ancient Hindu golden age.** India equal to (or better than) Greece and Rome — gave the world zero and "Arabic" numerals. One enthusiast said: *"The Asiaticks had climbed the heights of science before the Greeks had learned their alphabet."*
2. **Medieval decline.** Blind faith in old texts; "tradition replaced observation", "religion supplanted science". Blamed on rigid caste, and on **Muslim rule after AD 1100**. The Mughal break-up after 1707 = "anarchy".
3. **British rescue.** British rule brought back science from "centuries of decline".

> **WHY THIS STORY MATTERED**
>
> It **justified British rule**: "You fell, we came to save you." And it was not only the British who believed it — **many Indians accepted it too**. Some Indian writers still say Indian science hit its "lowest ebb" between the 12th and mid-19th centuries.

(c) The new "revisionist" view (Arnold supports this)

- **The Muslim period was creative, not destructive.** Example: medicine — **Ayurveda** (Hindu) and **Unani-tibb** (Graeco-Arabic) mixed in a "creative synthesis"; *vaids* and *hakims* did not fight each other.
- **India was not isolated.** It borrowed from and gave to the Middle East, Central Asia, China and South-East Asia — in agriculture, architecture, astronomy, chemistry, medicine, metallurgy, textiles, shipbuilding and arms. (Like Needham showed for China.)
- **Europe did not hit a "stagnant" India.** After 1498 (Portuguese arrival), India also gained from Europe, e.g. in shipbuilding and horticulture.
- There was a gap between **artisans** (craft skill) and **scholars** (literati) — but Europe and China had the same gap then. At **Akbar's court**, war and curiosity brought them together.
- **The 18th century was not "dark":**
  - Raja **Jai Singh's observatories** (1722–39) at Jaipur, Delhi, Mathura, Ujjain and Benares.
  - **Serfoji**, last Maratha ruler of Tanjore — library of Indian and Western medical texts.
  - **Hyderabad** and **Lucknow** (Islamic science, Unani), **Benares** and **Nadia** (Hindu learning), **Delhi** — a "twilight renaissance" until 1857.
- **Printing** — Jesuits brought the press to Goa in the mid-1500s, but it had little effect until the late 1700s. Arnold: this does **not** prove "mechanical backwardness". India adopted Western things **selectively**; the manuscript tradition and scribes had prestige. (Printing later exploded in the 1800s.)
- **War technology was adopted fast** — Mughals, **Tipu Sultan** (till 1799), and **Ranjit Singh**'s foundries at Lahore and Amritsar made heavy guns (1820s–30s).

(d) Diversity — a strength and a weakness

- India is huge and varied, so **many regional centres** flourished at once (regional schools of Ayurveda and Unani; regional weaving and dyeing). If one declined, another survived.
- But this same size and diversity made it **hard for anyone** — British rulers *or* nationalist scientists — to use science to "conquer and integrate" India into one unit.
- **Cities:** the new colonial ports (Calcutta, Bombay, Madras) grew with artisans and intellectuals who came from older centres. Old centres like Benares and Delhi survived. Universities with strong science by the 1940s (Lahore, Lucknow, Allahabad, Dacca) were in cities that were already important 200 years earlier.
- **Social groups:** old elites became agents of the new science — Brahmins, Vaidyas, Kayasthas became the Bengal **bhadralok**. India's only science Nobel before 1947 went to a Tamil Brahmin, **C. V. Raman** (1930). But Parsis, Indian Christians and lower castes also joined.
- **Princes** were big patrons — India's first major hydro-electric scheme was in princely **Mysore (1898)**, and the **Indian Institute of Science** came up in Bangalore about ten years later. Princes were often *more adventurous* than the British.

ARNOLD'S POINT ABOUT INDIAN TRADITIONS

India was **not a "tabula rasa"** (blank slate). Because pre-colonial Indian science was open and adaptable, an **interactive model** explains the colonial period better than either (a) a clash between two rigid systems, or (b) an automatic Western victory. **But** don't overdo it — colonial rule did cause a **"profound rupture"** (a deep break), both material and intellectual.

1.3 "Colonial science": Basalla and his critics

Earlier histories told science in British India as simply the **introduction and spread of Western ideas**. Indian traditions appeared only as "superstitious opposition" or as already dead. Now historians see it as more complex and two-way. To understand this debate, start with Basalla.

George Basalla's three-phase model (1967) — "The Spread of Western Science"

PHASE	WHAT HAPPENS
Phase 1 Contact & collection	Europeans arrive (trade, conquest). The "non-scientific" land is just a **source of data** — maps, plants, animals, minerals. Main sciences: **botany and zoology**, then astronomy, geology, geography. Only Europe (with a "modern scientific culture") can "appreciate, evaluate and utilise" this data.
Phase 2 "Colonial science"	Local scientists (Basalla means *Europeans* living there) and local institutions appear. But they are **dependent** on Europe — trained there, follow its agenda, can't easily enter top societies or the "invisible colleges" where new ideas are debated. Basalla said dependent ≠ inferior. He applied it even to the USA and Japan.
Phase 3 "National science"	Enough local scientists; **national institutions and honours**; science serves national needs. The USA and Russia reached this between the two World Wars.

What Basalla got right: botany and zoology really did dominate early colonial India (and this lasted into the 20th century, even though Indians tried to push towards maths, physics and chemistry).

CRITICISMS OF BASALLA — LEARN THESE 7 POINTS

1. **Ignores medicine and technology** — which mattered more than "pure" science in 19th-century India.
2. **Treats all colonies the same.** Settler colonies (Australia, North America — natives pushed aside) are very different from "colonies of exploitation" like India (few whites, strong local culture).
3. **Assumes "colonial science" = Europeans only.** Indians practised it too. (And not only British: Danish botanists, German foresters, American malaria experts worked in India.)
4. **Assumes science is "value neutral, objective … universally valid"** and that its spread is good and inevitable — Michael Adas calls this a "developmental teleology".
5. **Assumes colonial governments happily spread science.** Often they **distorted or blocked** it.
6. **Eurocentric** — says modern science could start only in the West; ignores India's and China's long traditions; makes non-Western people **passive**.
7. Even after independence, many ex-colonies **couldn't build "national science"** and stayed dependent on a few scientific superpowers.

Other arguments against "diffusion"

- **The receiving culture is active.** It "subverts, contaminates, and reorganises" the science that Europe brings (Raina & Habib). *Caveat:* before 1947, colonial political and financial control limited how far this could go.

- **Western science depended on local knowledge** — "local savants, scribes, interpreters and artists, fishermen and forest-folk". Many so-called Western "discoveries" in botany, geology and the trigonometrical surveys used Indian information.

An alternative: forget "centre vs periphery"?

- Some argue science was always **cosmopolitan**; the "metropolis" can move around (Roy MacLeod's **"moving metropolis"**).
- Colonial scientists were **equal participants**, not inferior; there was no real "low science" (fact-collecting in colonies) vs "high science" (theory in Europe) (Satpal Sangwan).

ARNOLD'S REPLY (VERY IMPORTANT)

This view rescues colonial scientists from being looked down upon — but it **ignores the "political character of science"**. India was a **colony**, not independent. Under both Company and Crown there was a clear **hierarchy**: Britain and its scientists at the top; India (and even British scientists working in India) dependent and subordinate. Even near the end of British rule, British authority in science stayed strong. As R. K. Kochhar says, Basalla's model **"trivialises the compulsions of colonialism"**.

Also, science is part of a **global hegemony** — even Nehru's India found it hard to get "scientific salvation". And inside India, science shaped **class** (new industries, new professional and working-class identities, new elites).

Arnold's conclusion: keep the term "colonial science", but use it **flexibly** — to describe "the various **technologies of power** operating within and through science in a colonial setting".

1.4 Science as modernity

Science was closely tied to both British and Indian ideas of what it means to be **modern**.

BRITISH SIDE

- At first (late 1700s) the British were **fairly open** to Indian ideas.
- By the early 1800s, science, technology and medicine became their **"civilising mission"** — proof of superiority over a "superstitious and backward" India. Science **increased the sense of difference**.
- After the Industrial Revolution, **steam technology** (steamships, railways) became the measure of civilisation (Adas, *Machines as the Measure of Men*). Also: Western vs indigenous medicine; belief that only the West could master India's environment.
- **Peak:** Lord Curzon (1899–1905) and "imperial science".

INDIAN SIDE

- Many Western-educated Indians **embraced** modernity. Gyan Prakash: **"scientific reasoning became the organising metaphor"** of the elite — they judged religion and society by science.
- "Colonial modernity" reached education, health, home life, gender roles, religion and social reform.
- But Indians **differed widely** in their responses.

The contradictions in British faith

1. Indians were seen as **"unready (or unfit)"** to receive scientific modernity.

2. The British wanted to **deny India** the competitive advantages full access to science and technology would give.
3. A **romantic anti-industrialism** — India should stay "a land of princes, peasants and artisans", spared ugly industry.

The problem for Indians: modernity = the West?

- If "modernisation" means accepting Western institutions and values (Lucian Pye's definition: advanced technology, the spirit of science, rationality, secularism, the nation-state), Indians will **always be "one step behind", "second-best and imperfect copies"**.
- **Partha Chatterjee:** Indian nationalism **challenged British political rule** but **accepted the same idea of "modernity"** on which that rule was based. (His book asks if nationalism was a "derivative discourse".)

Two ways out of this puzzle:

1. **Translation** (Prakash) — Indians did not accept science exactly as the West gave it; they **"translated"** and renegotiated it to fit Indian idioms and needs.
2. **Multiple modernities** — modernity is not one Western "patent"; it can take many forms, mixing Western science with Indian traditions.

Indian scientists did try to build their **own Indian modernity**, especially by **selectively using (or re-inventing) Hindu ideas** — with different amounts of "hybridity" in different people.

> HOW IT ENDED: NEHRU VS GANDHI
>
> **Nehru:** a worshipper at "the shrine of science"; science = both the tool and the symbol of Indian modernity. **Gandhi:** rejected modernity, "its alien sciences and inappropriate technologies". After 1947, under Nehru, a **pragmatic compromise**: tradition got only a secondary role; the state chose an internationally recognisable science and technology.

1.5 Chapter 1 in a nutshell

> THE MAIN POINT ARNOLD WANTS US TO GET
>
> 1. Science is **social and political**, not neutral.
> 2. India had **rich, plural, adaptable traditions** — the Orientalist "golden age → decline → rescue" story is wrong and served colonial interests.
> 3. The relationship was **interactive, not one-way diffusion** — but it was **unequal**, because of colonial power. So reject Basalla, but don't swing to the other extreme of "no centre, no periphery".
> 4. Science became the core of **modernity** — used by the British to claim superiority and by Indians to claim equality and nationhood. This created dilemmas (copying vs translating; Nehru vs Gandhi).

> EXAM TIP
>
> Almost every essay can start with: *"As David Arnold argues, science is 'a highly social activity' that cannot be 'sealed off' from the society in which it is practised. In colonial India, science, technology and medicine were shaped by Indian traditions, colonial power and competing ideas of modernity…"* And for any centre–periphery or "colonial science" question, use the **Basalla model + 7 criticisms + Arnold's reply**.

CHAPTER 2

Science under the Company

Book pages 19–56 · Period: roughly 1757 (Plassey) to 1858 (end of East India Company rule)

In one line: The East India Company did not "rule by science". It used science **when it brought profit, military advantage or prestige**, and otherwise left it to its army engineers and surgeons as a spare-time hobby. Arnold calls this a **"fitful flirtation"**.

2.1 The Company and modern science grew up together

The English East India Company (EIC) was founded in **1600** — the same time as the **Scientific Revolution**. Look at how its history runs side-by-side with big moments in science:

SCIENCE	COMPANY / BRITISH POWER IN INDIA
Royal Society of London, **1662**	Company already trading at Surat, Madras, Masulipatam
Buffon's *Histoire naturelle*, 1749	Anglo-French war in the Carnatic
Linnaeus's *Species plantarum* (binomial naming), **1753**	**Plassey, 1757** — British power in Bengal
Geological Society of London, 1807	British control over much of India; Marathas about to fall (1818)
Lyell's *Principles of Geology*, 1830	Company loses its trading rights, 1833
Darwin's *Origin of Species*, **1859**	Company abolished, **1858**

So the Company was "well placed" to use and support science. Arnold asks: **How central was science to the Company? Was it decorative/symbolic or really useful? Who did science and why? How was India shown through science?**

2.2 The Company as patron: control, stinginess, self-interest

Control and gatekeeping

- The **Court of Directors in London** had a "commanding position". The Company had a **near monopoly** over Western science in India and strictly controlled who could enter India.
- Outsiders were rarely allowed. Exceptions: French naturalist **Victor Jacquemont** (died in India, 1831) and botanist **Joseph Dalton Hooker** (1848–50). **Alexander von Humboldt asked but was never allowed.**
- Big British scientists like **Banks, Lyell and Darwin never visited India**. They depended on networks of Company army officers, surgeons and officials to send specimens and information. (Darwin's India informants included surgeons Malcolmson, Royle and Falconer, and museum curator Edward Blyth.)
- Through its control of jobs in India, the Company controlled one of the richest fields of scientific employment for Britons. Even inside the Company, scientists depended on the Directors' or Governor-General's approval and money.

Stingy and erratic funding — examples

- **Roxburgh's *Plants of the Coromandel Coast*:** Company paid £300 in the 1790s; by 1819 it cost £2,000 and the market had collapsed — a **"commercial disaster"** that made the Company less generous later.
- **Siwalik fossils:** Falconer and Cautley shipped **40 tons** of fossils to London. The Company agreed only to buy 40 copies of the catalogue at a guinea each (≈ 40 guineas) and refused the full cost; the British government gave £1,000.
- **Hooker's *Flora Indica*:** only one volume came out (1855); full publication had to wait till after the Company ended.
- In public, scientists praised the Company's "princely" patronage. In private, they called the Directors **"cheese-monger Emperors"**, "scoundrely" and "knavish".

Was it all about money? Two views

RAY DESMOND

The Company was not *only* commercial — it kept an unprofitable library and museum in London, supported scholarly books, and allowed officials to do Oriental studies.

ARNOLD

The Company was often attacked for misrule, and science was prestigious among Britain's ruling class. So being called "most munificent patrons of science" was **politically useful**. "Self-interest and scholarship often coincided" — but **self-interest was frequent; scholarship got only erratic support**.

- **Lord Wellesley (1804)** said promoting science was a "duty" of the British government in India. But (as Marika Vicziany shows) he knew surveys could **boost his reputation and deflect criticism** of his wars of expansion.
- **Anton Hove (1787)**, a Polish botanist sent by the Board of Trade, went to Gujarat **disguised as a poor physician** to report on cotton and textiles — clearly **commercial and political spying**.

2.3 The great surveys: Mackenzie and Buchanan

- **Colin Mackenzie** — survey of Mysore, **1799–1810** (after Tipu's defeat). Aim: a clear knowledge of the "extent, properties, strength and resources" of the territory, plus "objects of natural history".
- **Francis Buchanan** — Madras through Mysore to Kanara (1800–1), and Bengal & Bihar (1807–14). His orders covered **everything**: land, soil, rivers, weather, history, plants, animals, minerals (especially those used for food, medicine, trade), forests, mines, fisheries, agriculture, people, customs, trade. He was to send **"useful or rare or curious"** plants to the Calcutta Botanic Garden.

WHAT DO THESE SURVEYS TELL US?

- The phrase "useful or rare or curious" shows the **mix of the useful and the ornamental** in Company science — but it was the **useful (material worth)** things that were asked about most.
- Buchanan himself loved botany and the idea of **"improvement"** (from Sinclair's *Statistical Account of Scotland*). He also had Romantic moments about waterfalls and forests.
- Arnold: these surveys are early examples of **"governmentality"** (Foucault) — the state collecting information on wealth, population, resources and customs in order to govern.
- The surveys served many purposes; the motives of individuals (Mackenzie, Buchanan) were not always the same as their paymasters'.

2.4 Mostly a spare-time, "gentlemanly" science

- The surveys were **exceptional**. Usually the Company **did not start** scientific projects; it just **tolerated** its employees' private hobbies — keeping weather records, hunting fossils, noting plants and animals.
- Official duties got in the way. In 1788 it was said that "a mere man of letters … allotting his whole time to philosophical or literary pursuits" was "a character unknown among Europeans resident in India".
- Unlike Britain, India had few European clergymen, landlords or manufacturers to write papers or fund societies. Scientists like Roxburgh lacked time. Many **died young or went home**. B. H. Hodgson was a rare naturalist who stayed on.

> *"Far from embodying the rule of science in the service of empire, the Company's involvement in science often appeared more like a **fitful flirtation**… The British enjoyed the company of science; it would be excessive to suggest that they ruled by it."* — Arnold, p. 25

- Much of the science before 1858 belonged to the British tradition of **"gentlemanly" (amateur) science** — no special state agency, no clear agenda. The government was even uninterested in an Indian branch of the British Association for the Advancement of Science (1830s).
- **Army engineers** (like Mackenzie) and **surgeons** (like Buchanan) gave "all-purpose expertise" — cheaper and faster than bringing specialists. Only late did specialists come: **Thomas Oldham** (from the Geological Survey of Ireland) came in **1851 → Geological Survey of India (1856)** — and was given only "an office, a clerk and a chaprassi"!
- The main exception: the **Great Trigonometrical Survey (1818)** — but it too used army men.

2.5 How colonial science was organised

Science on the march — and the Siwalik fossils

- Early geology was written by army officers and surgeons **during troop marches** — Captain Franklin in Bundelkhand (1826–7); Assistant Surgeon Hardie slipping away from his regiment to look at rocks.
- **Siwalik fossils:** first found by Lt. **Proby Cautley** and his assistants while digging the **Jumna Canal** (1834). Surgeon **Hugh Falconer** followed up in Nov 1837 — and a **local raja** who owned a huge mastodon tooth ("the tooth of Deo") told him where to dig: **300 fossil bones in six hours**. (Indigenous knowledge!)
- Falconer and Cautley won the Geological Society's **Wollaston Medal** (1837). Falconer became FRS (1845) and a big name in British geology. India could be a **"springboard"** for a career in London (also George Everest).
- But **Lyell**, giving the medal, praised them for overcoming being "not versed in fossil osteology" and stationed on "the remote confines" — i.e. colonial science = **amateurish**. Falconer himself hated going back to "isolation".

Isolation → a "diffident" and "dependent" science

- Scientists in India saw themselves as **amateurs** whose job was "collecting and recording with precision" — sending **raw data** for expert minds in Europe to theorise. Franklin: "I am not learned enough… to become an advocate for any party."
- **James Prinsep** wrote 60+ articles (1825–39); could have been "the Humboldt of the east" but died at 41.
- **Oldham (1856)**: "We are working under the same difficulty that would affect a tradesman without his tools or a physician without medicines."

Societies, museums, journals

- **Asiatic Society of Bengal** — founded **January 1784** by Sir **William Jones**, modelled on the Royal Society. Mainly literary/linguistic, but it was the **main platform for Western science in India until the Indian Science Congress**

(1914). Its "Physical Class" (1808, revived 1828) met erratically. Other bodies: Agricultural Society (1820), Medical and Physical Society (1823), societies in Bombay and Madras; Royal Asiatic Society in London (1823).

- **Museums:** Asiatic Society museum (proposed by Nathaniel Wallich, 1814); Bombay Literary Society museum (1815); a museum of economic geology (1835, for coal) → given to the Geological Survey in 1856. **Edward Blyth**, full-time curator 1841–62, was so tied down he "hardly [had] been out of" Calcutta in 14 years.
- **Journals:** *Gleanings in Science* (1829, Capt. J. D. Herbert) — filled with local items from steamboats on the Hooghly to **keeping wine cool**! *Calcutta Journal of Natural History* (1840, John M'Clelland) — "it would not pay" — folded 1847.
- **Public apathy:** Geological Society of London had 700+ members; Asiatic Society usually under 250 — just **119 in 1845**. Bengal's journal-reading public was under 2,000 (1829).

Science as a badge of the European elite

- Science helped the British in India feel **connected to cultured Europe** and **different from** both Indians and lower-class Europeans (soldiers, merchants, planters) — Matthew Edney's point.
- The *Calcutta Review* (1844) said "the bane of this country is ignorance" — not in "the dark recesses of native life" but "in high places" — i.e. criticism of the Company for not caring about science.

"About India, not for Indians" — the externality of science

- **Bentinck (1835)** said the "great object" was to promote European literature and science among Indians. But in practice Western science was **"primarily about India and Indians, and only secondarily for and by them"**.
- The Asiatic Society had **no Indian members until 1829** (later Ramkamal Sen became "native secretary", 1838). The Bombay Literary Society admitted Indians only after 1840 — a "departure from the original design".

> **BUT INDIANS WERE NOT JUST SPECTATORS**
>
> - **Bal Shastri Jambhekar** — Brahmin from Ratnagiri; at 17 became secretary of the Bombay Educational Society (1827); translated Brougham's *Treatise on … Science* into Marathi; professor of mathematics at **Elphinstone College (1834)**; ran the Anglo-Marathi journal *Darpan* to spread "useful" Western knowledge.
> - **Calcutta's Agricultural and Horticultural Society** had Indian members from the start — **Dwarkanath Tagore**, **Radhakanta Deb** — and published some work in Bengali.
> - Middle-class Indians in Calcutta and Bombay were attracted by the **practical uses** of Western science and wanted to know **how it gave the British such power** over the material world.

2.6 Science and Orientalism: from respect to rejection

- **C. A. Bayly:** the "colonial information order" was **built on Indian foundations**; Indian knowledge systems survived "traumatised, but recognisably the same". We must recognise "Indian agency".
- **Jones's 1784 agenda** for the Asiatic Society: study everything about "man and nature" in Asia — geography, history, astronomy, maths, medicine, chemistry, agriculture, music, even pottery.
- **Thomas Trautmann:** this was an attempt to **fit India into Western knowledge** (e.g. Jones's discovery that Sanskrit, Greek and Latin are related; matching Hindu legends with Biblical chronology). Orientalism looked for connections, but ended in **difference and a hierarchy of nations**. India's knowledge of itself was never "adequate"; Jones even said in science "Asiatics" were **"mere children"**.

Case 1: Astronomy

- Very **practical** for the British — finding longitude for navigation and maps. Madras Observatory's Michael Topping: astronomy is "the parent and nurse of navigation".
- Observatories built 1792–1842 at Madras, Calcutta, Lucknow, Trivandrum, Poona (by the Company or Indian allies). Europeans worked at Indian courts (J. D. Herbert at the Nawab of Awadh's Lucknow observatory, 1831).
- At first astronomy looked like a **bridge**: Jai Singh's observatories were admired; Colebrooke wrote sympathetically; **John Playfair (1789)** said Indian astronomy was a separate, very ancient and accurate system. But Trautmann: the question was never whether Indian astronomy could teach Europe something — Indian astronomy was judged using **European astronomy as the standard**.
- Later, **contempt**: *Calcutta Review* (1844) said Hindus had "retrograded", the *Surya Siddhanta* was only 11th–13th century, and the "Puranic system" was "a mass of absurdity" to be "demolished" by "the measuring-rod, the theodolite and the telescope". **Macaulay (1835)**: Indian astronomy would "move laughter in girls at an English boarding school".

Case 2: Botany and plant names

- **Linnaeus** made botany a project of rationalisation — one standard system (binomial names) to name plants worldwide.
- **Jones** wanted Indian plants to keep their Sanskrit names and botanists to learn Sanskrit. Ignored. Roxburgh used a "motley" mix of languages; Buchanan recommended Sanskrit but wasn't consistent.
- Buchanan complained villagers used vague names ("large", "small", "wild", "cultivated"), many names for one plant. Robert Wight got half a dozen names for the same plant within a few miles. Risky for medicinal or poisonous plants.
- So, increasingly, **"the testimony of nature was preferred to the testimony of 'natives'"**. Hooker — who relied on "a small army of local assistants" — still called Indian botany "a perfect chaos of new names". (Exception: **Brandis** in 1874 urged foresters to learn Indian names because they were more fixed.)
- **Result:** Indian plant naming moved away from Sanskrit and **conformed to European taxonomy** — a sign of a wider trend towards **conformity** in colonial science.

2.7 The Great Trigonometrical Survey (GTS)

- Before: **James Rennell**, first Surveyor-General (1767) — *Bengal Atlas* (1780), *Map of Hindoostan* (1783). But mapping was haphazard; distances were guesses.
- **Triangulation** came from Europe (British Trigonometrical Survey 1784, Irish Ordnance Survey) — a **metropolitan precedent**.
- **William Lambton** (infantry officer) began in **1799** using 100-foot steel chains; Madras to west coast, finished 1817. **GTS founded 1818** with Lambton in charge.
- **George Everest** re-measured with "Colby's compensation bars" from Britain and by 1843 measured the **"great meridional arc" — 1,500 miles from Cape Comorin to Dehra Dun**.
- In technique and scale it was **ahead of any survey in Europe** → colonial India could **generate its own techniques**, not just copy. It was in effect **India's first dedicated scientific service**.

WHY THE BRITISH WANTED IT

- **Military**: maps for war; it grew directly from **Tipu's defeat (1799)**; later Punjab, fear of Russia.
- **Navy & trade**: exact locations of ports and coasts.
- **Political**: turn diverse India into "a single, mapped entity, an ordered and rationalised imperial space".
- **Scientific prestige**: measure the true shape of the earth.

HISTORIANS' VIEWS

- **Cannon:** it shows British policy "better than still another study of Macaulay's education minute"; "overshadowed all European geodetic operations".
- **Edney:** survey as a "panopticon" to "see" and "know" India — but a **"flawed geographic panopticon"**, an **"empiricist delusion"**. It never covered all India (cost, terrain, illness, small staff).
- **Arnold:** still remarkable that a "cash-conscious" Company supported it for decades — few projects were so consistently backed.

INDIAN ROLE IN THE SURVEY

Heavy reliance on **Indian skill** — e.g. **Radhanath Sikdar** (from 1831) and the "pandits" who secretly surveyed Tibet after 1860. But Indians were kept in **subordinate posts**. Edney: **"By measuring the land… the British distinguished themselves from the Indians: they did science, the Indians did not."** The survey turned "mystical, religious, Hindu space" into a "rational, scientific, imperial structure of space" (p. 43).

2.8 Geology and zoology

- **Zoology** stayed amateur — not obviously "useful", no official base; driven by collectors and **hunters (shikar)**. Jerdon's *Mammals of India* (1874) was written for "sportsmen and observers". Zoological Survey only in 1916.
- **Geology** was slow at first because:
  - The Company and London pressure groups **didn't want India to compete** with Britain's copper and minerals.
  - India was supposed to stay an **agricultural** land.
  - Ignorance of resources, lack of experts; jungles, malaria, tigers ("tigers have no regard for science"); no deep cuttings until canals and railways; India's rocks seemed "monotonous".
- Slowly it became exciting: **kankar**, **laterite** (Buchanan, 1800), deep alluvium under Calcutta, the **Deccan lava flows**, the Siwalik fossils, the Himalayas.
- **Steam changed everything.** Steamships (and later railways) needed **coal → Coal Committee 1836**. But slow — M'Clelland (1844): "the government move slowly in matters of improvement". Oldham (1851) → GSI; **27 coalfields** examined in 15 years. Geology moved from **amateur hobby to a branch of the state**.

2.9 Botany: profit + philanthropy

- Botany was the **most useful** and the **least "provincial"** Company science — closest to metropolitan science. The Company traded in spices and indigo, and its servants were always sick (medicinal plants).
- **J. G. Koenig** (Danish surgeon, 1778) — first naturalist employed by the Company; brought **Linnaean taxonomy** to India. (Earlier: Garcia d'Orta in Goa, 16th c.; Van Rheede in Malabar, 17th c.)
- **Calcutta Botanic Garden** — proposed by **Col. Robert Kyd (1786)**, approved **1787** with Banks's support. Two motives:
  - **Famine protection** — new food plants (sago, date palm) after Bengal's famines.

    - **Profit** — grow cinnamon (a Dutch monopoly), indigo, cotton, tobacco, coffee, tea, to "**outstrip our rivals**".

  Arnold: a "judicious blend of **profit and philanthropy**" — commercial self-interest in **humanitarian guise**.
- Under **Roxburgh** (classified 2,000+ species) it became more about collecting and classifying than improving Indian farming.
- **Saharanpur garden** (proposed 1817) — **J. Forbes Royle** (1823–31) tested cotton, tobacco; tea succeeded, cinchona failed; for "the comforts of the people" and "the wants of a great empire".
- **Other meanings:** missionary **William Carey** saw botany as praising God ("All Thy Works Praise Thee, O Lord"); botany as **art** — Indian artists **Vishnu Prasad** and **Laksman Singh** painted plants.
- **Botany + medicine:** of 42 botanists before 1840, **28 were surgeons**. The medical service was an "all-purpose scientific agency". The Company searched for **local substitutes for imported drugs**. **Royle** studied bazaar drugs and argued Hindu medicine was **older** than Graeco-Arabic medicine (1837). Later he became a professor in London — another colonial career → metropolitan career.

2.10 Science in the tropics: India as "tropical"

- Science also placed India in the natural world — finding links with Europe (universality) but also **differences**. The idea of **"tropicality"** became a central paradigm: India as a site of tropical plants, animals and **diseases**.
- The idea came from the **West Indies**: tropics = unfit for white settlers, good for slavery and plantations.
- India did not fit at first — it was a land of **famine** (Bengal 1770, ~10 million deaths), not tropical abundance. But by the 1820s–30s "tropical" was common in texts, as the earlier respect for Indian civilisation faded and India was seen as **vulnerable to a cruel nature** — needing **European science to "redeem"** it.
- **Joseph Banks** saw India as a **tropical estate** for Britain — sugar, cotton, coffee, indigo; in 1788 he even predicted tea could grow between Bengal and Bhutan.
- **Royle** (inspired by Humboldt) divided India into **tropical, temperate and alpine zones** — so almost any useful plant could grow somewhere in India.
- The **Himalayas** became a "spiritual homecoming" for Europeans — oaks, pines, violets reminded them of home. **Hooker** called **Darjeeling** ideal for a sanatorium — a precedent for later **hill-station research institutes**.

> **BIG SHIFT BY THE END OF COMPANY RULE**
>
> Jones and the early Orientalists saw India as an **ancient (if decayed) civilisation**, like Greece and Rome. Later scientists turned away from Sanskrit texts and pandits to **direct observation of nature** — and saw nature as a tyrant over races "bereft of modern science and technology". It became "axiomatic" that **without Western science there could be no "improvement" in the tropics**.

2.11 Chapter 2 in a nutshell

WHY THE COMPANY ENGAGED WITH SCIENCE	WHY ITS SUPPORT WAS WEAK / ERRATIC
• Revenue & trade (botany, cash crops, coal) • Military (maps, surveys after Tipu & Marathas) • Knowing new territories to rule them (surveys) • Prestige & deflecting criticism (Wellesley) • Famine (botanic gardens) • Health of its servants (medicinal plants)	• Science was secondary to revenue, law and order • Cash-conscious Directors; "commercial disaster" of Roxburgh's book • Fear of Indian competition (minerals) • Scientists busy with official duties; few, isolated, died young • Small, apathetic public

THE MAIN POINT ARNOLD WANTS US TO GET

1. The Company's relationship with science was a **"fitful flirtation" — opportunistic and utilitarian**, not a planned "rule by science".
2. Science was mostly done by **army engineers and surgeons** as **gentleman amateurs**, who sent raw data to Europe — a **dependent, isolated** science (but sometimes **ahead** of Europe, as in the GTS).
3. Science also worked as **identity and prestige** for the European elite, and it was **"about India, not for Indians"**.
4. Indian knowledge and helpers were **essential** (raja and fossils, Sikdar, plant names, artists, informants) — but kept **subordinate**, and over time European science **moved away from Indian knowledge** towards conformity with Europe.
5. By the end, science framed India as **tropical, backward and needing Western rescue** — and practical needs (coal, maps, steam) were pushing the state towards a more professional science (GSI 1851/56).

CHAPTER 4

Technologies of the Steam Age

Book pages 92–128 · Textiles, mining, ships & steamships, roads & railways, telegraph, irrigation, ideology, Swadeshi & Gandhi

In one line: Steam-age technology did not simply "transfer" from Britain to India. It was a **"dialogue"** shaped by **colonial power** (the British pushed what helped them and **blocked** what didn't), by **Indian conditions, labour and older technologies**, and it carried **mixed meanings** — progress for some, domination or ruin for others.

4.1 Arnold's opening argument

- The usual story: the 19th century saw "a massive transfer of technology from the West to Africa and Asia", with colonialism as the channel — technology as a **"tool of empire"** (Daniel Headrick). India is the classic example.
- Arnold: this is **too one-dimensional**. It praises Western dynamism but ignores the **context**. Technologies are rarely moved "wholesale". Transfer is more like a **"dialogue"** (Arnold Pacey) — especially in India, which already had many technologies and a very different environment.
- **Colonialism shaped the dialogue:**
  - It **cut India's old "inventive exchanges"** with Asian and Indian Ocean neighbours and made Britain almost the **only partner**.
  - The state **promoted** technologies that served its **military, economic or ideological** needs and **restricted** Indian access to technologies that could hurt British interests.

> *"**Denial was often as important as diffusion** in shaping the technology of colonial India."* — Arnold, p. 92

- **Old and new lived side by side** — in farming, textiles, transport and labour. The relationship was often **"synergistic"** (helping each other), not just a fight.
- **Mixed messages:** steam meant "progress" and "empire" to the British, but not necessarily the same to Indians. Both Europeans and Indians had **doubts** about whether modern technology suited India.

4.2 Textiles: decline, adaptation and survival

The old nationalist story: "de-industrialisation"

British mills first took India's export markets, then flooded India with cheap cloth. Weavers lost work and were pushed into farm labour → more poverty → famines. India's own mills (after 1850) could not absorb them.

What new research adds

- **Imported machines, dependence:** over **90%** of textile machinery for Indian cotton and jute mills (1850s–1930s) came from **Britain (Lancashire)** — worth Rs 780 million in 1905–39 alone. India barely developed its own machine-making before 1939.
- **But local practices shaped mills:** the **jobber** recruited and managed workers (bridging managers and workers); **Julahas** (Muslim weavers from north India) worked in Bombay mills.
- **Pre-colonial India:** probably the world's greatest cotton-textile producer until the late 1700s; the **charkha** (maybe from Iran, 14th c.); block printing, dyes, mordants; European "craze" for Indian calicoes.

- **But its strength was labour, not machines:** cheap labour, plenty of cotton, and the skill and "hereditary knowledge" of spinners, weavers and dyers. Bengal looms were basic (no treadle); **20 stages** from raw cotton to cloth, each by a caste specialist. Quality was "inversely related to technical sophistication" — the finer the cloth, the more labour. Weavers had **no capital** and depended on middlemen. So India was **not about to have its own industrial revolution**.
- The Company supported **silk** filatures (to compete with Italy) but showed **no interest** in bringing cotton-spinning innovations to India. With no tariff protection, British cloth won: Indian cloth exports fell from **Rs 165 lakhs (1816–17) to Rs 8 lakhs (1830–31)**; imports rose from Rs 3 lakhs to Rs 60 lakhs. Steamships, the Suez Canal (1869) and railways made it worse → **new technological dependence on the West**.

HANDLOOMS ADAPTED AND SURVIVED

- Village weavers disappeared; survivors moved to **specialised urban centres** (e.g. Salem, Sholapur) making **ceremonial cloth, dhotis, saris**.
- Railways helped them sell wider; Indian migrants overseas created markets for handloom **lungis**.
- Used cheap **mill yarn**; adopted the **fly-shuttle** and **jacquard loom**; after 1920, **electric power**.
- In the 1930s, handlooms still made about **one-third** of India's textiles. "Dynamics of continuity" — not passive survival.

Changing image: at first Europeans wanted to *learn* from Indian textiles (Jones 1784, Hove 1787; Banks 1791 wanted to know "the minute practice of Indian workmen"). By the early 1800s (Buchanan) Indian tools looked "crude"; Indian crafts became the symbol of **backwardness**. By the late 1800s village crafts were nearly dead, sometimes seen as a cover for "criminal tribes and castes", and were put in the **ethnographic museum** as "relics of the past".

4.3 Mines and metallurgy

- Past glory: Delhi's **Iron Pillar**; **wootz steel** admired in Europe into the 19th century.
- Growing **disdain**: Valentine Ball (1881) said critics claimed Indian iron-workers "possess the art… [but] know nothing of the science" — though he admitted European foundries also lacked "scientific guidance". Fuel scarcity hurt too: **14 tons of charcoal per ton of iron**. Not all decline was the fault of British rule.
- Indian mining was seen as crude: no gunpowder, pit-props or pumps → short shafts that flooded or collapsed.
- **Examples:** Himalayan copper mines — a Cornish miner (1838) suggested steam pumps, but mines were abandoned (1841) because English copper was cheaper even inside India. **Kolar gold** — rediscovered 1802, reworked only after 1871, deeper after **electric power (1900)**. Rajasthan lead purer than British but one-eighth dearer. Panna diamonds — shallow pits, no investment.
- **Iron:** Dharampal estimates **10,000 furnaces** in 18th-century India, each ~20 tons a year — "wholly crushed out of existence" by British competition by 1880. India became a **market** for British iron (£15 million a year). The **Porto Novo** steel company (1833) made good steel but collapsed by the 1860s (charcoal shortage, transport costs).
- **Coal: Raniganj** (dug from 1777) — bought by **Dwarkanath Tagore** in 1836; 40+ mines by 1880s; two-thirds of India's coal was home-produced. But very **labour-intensive** — Santal and Bauri families with picks and crowbars; small mines looked like "badly made tanks". Electricity improved mines only in the early 1900s.

POLITICAL DENIAL — THE KEY POINT

Metals and metal-working had supported Indian kingdoms' **arms and cannon** (from Akbar to the Marathas and Sikhs). So the British **limited India's ability to mine and work metals**, just as the **Arms Act (1878)** limited Indian access to firearms. Rajasthan's mines (Tod's "mines of Mewar") were abandoned; in **1857** the **lead mines at Ajmer were closed** because lead meant bullets.

4.4 Shipbuilding and steamships

- India had an old shipbuilding tradition using **teak**. European trade first **helped** it: Indian-built ships were cheaper and tougher.
- **Hybrid ships** = Indian carpentry + European design. Indian contributions: **rabbeting** (planks fitted so closely that no caulking was needed), **coir ropes**, **lime** to stop worms.
- **Bombay Dockyard:** Parsi carpenters from Surat (1736); **Lowji Nusserwanji Wadia** became Master Builder — 35 ships in 40 years; his sons built 30 more. The *Britannia* (1778) impressed London so much that more ships were ordered; some joined the **Royal Navy**. 1736–1821: **159 ships** (15 over 1,000 tons). Called "vastly superior to anything built anywhere else in the world".
- **Decline after 1820**; last big Royal Navy ship 1831. The yard built teak hulls for imported engines (e.g. *Hugh Lindsay*, 1829, Bombay–Red Sea). Most steamships were **assembled from British parts** → British firms gained.

WHY DID INDIAN SHIPBUILDING DECLINE?

Headrick: technology — "what destroyed Indian shipbuilding was British iron".
Sangwan: politics — British shipbuilders **lobbied Parliament** to restrict Indian ships and cargoes, and feared steamships being *made* in India.
Arnold: the political point is valid, but the **technological gap** after industrialisation was also real.

CONTINUITY: PARSIS

Shipbuilding was a route for Parsis from crafts to modern science. **Ardaseer Cursetjee** — engineer, pioneer of gas lighting and photography in Bombay, Chief Engineer at the dockyard, the **only Indian elected FRS in the 19th century**. Later Parsis: **J. N. Tata**, **Homi Bhabha**.

- **Steam in the capitals:** Boulton minted Company coins by steam (1786); the **Calcutta Mint (1831)** made 300,000+ coins a day — steam serving the colonial state.
- **Ganges steamers:** Governor-General **Bentinck** came to India on the steamer *Enterprise*; iron steamers cut Calcutta–Allahabad from **3 months to 3 weeks** — vital for mail and government.
- **Limits:** rivers rise and fall with seasons, sandbanks, high costs. Worked best on the **Brahmaputra** (Assam tea, "coolie" labour). On the **Indus** (1837) a **"technological trap"**: a strong engine made the boat too heavy (ran aground); a light one couldn't fight the current. "So the steamboats died a natural death."
- **Symbolic meaning:** steamers were "conspicuous heralds" of Western science. In the novel *Oakfield* (1854) a character sees an "inconceivable separation" between the English "making a servant of the Ganges" and Indians "worshipping the same river". Arnold: but since many Indians quickly used steamers and trains, the gulf was **not as wide** as the British imagined.

4.5 Roads and railways

Roads first

- Pre-steam transport was not frozen since antiquity — Mughals built roads; carts, bullocks, camels, and **Banjara** pack-bullock traders.
- 1780–1840 the British built roads for **military** reasons: Calcutta–Benares road (1781); the **Grand Trunk Road** rebuilt from 1833 using pounded **kankar** ("Indian macadam"), reaching Peshawar after Punjab's conquest; Bhor Ghat road; roads in the south after Tipu's defeat. Like the Survey, roads **imprinted colonial power** on the land.
- **Lalla Tantimul**, an Indian contractor, started a horse *dak* (mail and passenger) service in 1849. Letters Calcutta–Benares: **2.5 months by boat → 52 hours**.
- British loved comparing themselves to **Rome** — but roads lagged. **Famine relief workers** replaced convicts as road builders — a labour-intensive, low-tech method.
- **Narmada coal experiment** (1838, 1848): Banjaras threw down the coal sacks "as they do grain" and said sharp coal would hurt their bullocks; boatmen lost boats in rapids; only half the coal reached Bombay after 8 months, **costlier than Welsh coal**. Lesson: inland coal needs **railways**.

Why railways? (the British motives)

1. **Coal** — one of the first lines (1855) ran from Hooghly to the **Raniganj coalfield**.
2. **Trade** — raw cotton from the Deccan; "intermesh" the economies of Britain and India: British goods in, Indian raw materials (cotton, oil-seeds, grain) out.
3. **Famine** — the **1880 Famine Commission** called for **20,000 more miles** as "protective" (anti-famine) lines. Under *laissez-faire*, controlling grain prices was rejected, but building railways was seen as legitimate. So famine pushed technology towards **agrarian needs, not an industrial base**.
4. **Size and security** — **Dalhousie's minute (April 1853)**: "immeasurable" political benefits — quick news of "every event", fast troop movement, fewer soldiers dying on long marches. MacGeorge: railways shrank India to **one-twentieth** of its size. (Though three different gauges didn't help!)
5. **Ideology** — Lord **Hardinge (1848)**: steam would be "the greatest instrument of civilisation for the people, and of strength for the Government". **Dalhousie**: railways are "great engines of social improvement". MacGeorge: railways "awakened the Eastern world" and broke "deep-seated ancient prejudices" like caste.

The reality check

- **Huge scale:** 40,000+ miles by 1947 (24,000 laid 1853–1900); 5th-largest system in the world by 1914; 600 million passengers a year by 1939.
- **But** the 1928 Royal Commission on Agriculture still called railways "backward"; they **relied on bullock-carts** to bring cotton to railheads and grain to villages.
- "A **British show**": Marx predicted railways would be "the forerunner of modern industry" in India — but most benefits went to Britain. Rails, sleepers, bridges, engines (even drivers) were imported. **14,000+ locomotives sold to India vs barely 700 made in India** (1850s–1940s); British pressure blocked Indian production. Workshops (Lahore: 10,000 workers) mostly did **repair and assembly** — though some call them "schools of skills".

NOT A STRAIGHT "TRANSFER" — INDIAN CONDITIONS AND INDIAN METHODS

- **Environment:** doubts in the 1840s about monsoons, floods, heat, animals; brushed aside by engineer Simms (1846) and Dalhousie — but easier said than done.
- **Rivers:** dry sand in summer, floods in monsoon → very long bridges. European piles failed; engineers learned the **Indian well-sinking technique** (sinking weighted brick/masonry piers). The Son bridge: 7/8 mile, 28 spans, £330,000.
- **Bhor Ghat incline**: 16 miles, 7 years, **42,000 workers** at peak (1861).
- **Labour:** artisan castes like the **Wudders** (tank- and well-diggers). Ian Kerr: engineers accepted Indian work practices because "the cheapness and relative abundance of Indian labour was worth many a machine". Customary tools (mattocks, crowbars, baskets) worked better than wheelbarrows or steam diggers. Indians learned new skills fast (riveting). Many women and children worked in family units.
- Construction became "routine" only by the **1890s** — 40 years of learning.

4.6 Telegraphs

- The clearest case of **military and political** motives. Grew alongside (often ahead of) railways — trains need telegraph signalling.
- Growth: 4,250 miles by 1856 → 17,500 (1865) → 52,900 (1900) → 100,000 miles and 17 million messages a year (1939). Unlike railways, telegraph **stayed a state concern**.
- Uses: news of the fall of Rangoon (1852); early warnings in **1857** ("the telegraph saved India" — an exaggeration, but it helped). Soon used by **businessmen and nationalist politicians** too. Submarine cable to Britain (1870) tied India into the empire.
- **Personal interest of governors-general** mattered: Bentinck (survey, steamers), **Dalhousie** the "technophile" (railways, telegraph, post), Curzon (imperial science).
- **W. B. O'Shaughnessy** (Company surgeon, chemistry professor) built a 21-mile test line (1839) **adapted for India** — thicker cables covered with cloth and pitch for heat and humidity. Later first Director-General of Telegraphs. His chief assistant was an Indian, **Shib Chandra Nandy**, who built the first line (Calcutta–Diamond Harbour) and lines to Allahabad, Benares and Dacca — before such posts were "entirely Europeanised".

4.7 Irrigation

- India had many water technologies: earth dams, step-wells, inundation canals, **anicuts** in Tinnevelly, and up to **40,000 wells per district** in Salem and Coimbatore. Some were decaying by the early 1800s (war, British neglect of maintenance, revenue pressure).
- But old works **survived**: in **1901, private works (tanks, wells) watered ~60%** of irrigated land; state canals only ~40%.

South India: continuity with Indian methods

- **Grand Anicut** on the Cauvery — "the greatest engineering work carried out in India before British rule".
- **Arthur Cotton** (Royal Engineers) built a new anicut on the Coleroon (1835–38) to save the Tanjore delta — added 250,000 acres. Then the **Godavari anicut** (1847–52) after the 1833 famine.
- Cotton found the Grand Anicut was made of **earth and rubble encased in stone** — and copied this cheap "rude mode of construction". This low cost helped get the project approved. He openly admitted his debt to the **"old native engineers"**. (He also believed irrigation would help spread Christianity.)

North India: new engineering, shaped by local conditions

- British first **restored Mughal canals** — Shah Jahan's Delhi canal (Capt. Blane, 1817–20), the Jumna Canal.
- After the 1837–38 famine: the **Ganges Canal** (1842–54) under **Proby Cautley** (the fossil-hunter!). Head-works at Hardwar; the huge **Solani Aqueduct**; dug by "basket-labour". A steam locomotive brought in (1851) soon broke down and ended up driving machines at Roorkee engineering college. Irrigated **1 million acres**; a model for the USA and Australia.
- It had defects and was remodelled (1866–70) — "shaped bit by bit to the circumstances of the country". So it was **neither a pure Western transfer nor an Indian copy** — an **"in situ" initiative**.
- **Army engineers** ran most canals, roads and survey works till 1900 — useful in peace, ready for war; they knew how to "deal with the natives". They saw themselves **conquering nature** like Romans ("battle with the river") — **technocratic paternalism**.
- By 1901, about one-fifth of British India's cropped area was irrigated (one-third of that by the state).

THE CANAL DEBATE

Critics: waterlogging, ***reh*** (salt crust), **malaria** from leaky canals (Dempster 1845; Corbett 1870s said famines got worse). **Elizabeth Whitcombe:** canals were "a costly experiment".

Ian Stone: negative effects exaggerated; canals reduced drought, peasants used them "rationally"; wells and canals were complementary.

OFFICIAL BLIND SPOT

The 1901–03 Irrigation Commission (set up by Curzon) **barely looked** at environmental costs. By the 1900s, focus shifted to **scientific agriculture**, and electric **tube wells** began to offer a more flexible way to irrigate.

4.8 Technology, ideology and resistance

Technology as the British "monument"

- Railways and canals were **monuments to British power and generosity** — the Raj as a **"technological empire"** that could master nature, end famine and turn "workshy peasants into model farmers".
- **After 1857**, direct interference in Indian society seemed dangerous, so **technology became the safer way to "improve" India**. Queen Victoria's Proclamation (1858) promised "works of public utility". The rebel **Begum of Awadh** replied that the Queen seemed to expect **"no better employment for Hindostanees than making roads and digging canals"**.
- W. W. Hunter (1862): railways would make British troops 20 times more efficient; commerce would be better security than "all the legions of England".
- **Deakin (1893)**: "If the British in India had achieved nothing else, the public works policy … would fully justify their supremacy". **MacGeorge**: no alien nation, "not excluding … ancient Rome", left more enduring monuments; better than "Mohammedan or Hindu creations" because they were useful.

"Indians hate technology"? — Arnold says be careful

- Sir John **Kaye** claimed railways and telegraphs "disquieted" Hindus in 1857 and made Brahmins look "as feeble and impotent as babes and sucklings". Cutting telegraph wires in 1857 was used as "proof" of Indian hostility to technology.
- **Arnold's response:**
  - Yes, rumours linked new technology to drought, disease and colonial coercion; railways did spread disease and ruin crafts. **Medical measures** (vaccination, plague controls) and **survey parties** (fear of new taxes) met real

resistance.

- But the British **exaggerated** resistance to project Indians as ignorant and themselves as rational.
- Destruction in 1857 was **strategic**, not superstitious. Rebels used **Enfield rifles** themselves; Bahadur Shah's proclamation promised merchants **free use of steam-vessels and steam-carriages**.
- Indians in Calcutta and Bombay **welcomed** the first railways; by 1858 the railway was "quite naturalised in the native mind". Trains carried **pilgrims** to melas; think of the train in Satyajit Ray's *Pather Panchali*.
- **Keshab Chandra Sen** (1866) praised the "wonders" of steam and electricity — but warned modern civilisation was "materialistic": "ill fares the spirit".

4.9 Steam, Swadeshi and Swaraj: is industry right for India?

- **Western critics of industry** (Ruskin, William Morris) admired Indian crafts.
- **George Birdwood** (1878 Paris Exhibition catalogue): in India everything handmade is "a work of art"; craftsmen "polluted no rivers… nor poisoned any air"; mill workers were like the grinder of a barrel organ. He mocked a **steam-plough** that sank in Deccan soil and was then "worshipped as a lingam" — praising the "Mahratta plough", the "product of three thousand years' experience". (Farm machines generally failed; experimental farms became "museums" of rusting tools.)
- **E. B. Havell (1901)**: India should not copy Western industrialisation (slums, class conflict); instead **revive the handloom** with improved **fly-shuttle looms**. Adopted by the Basel Mission, provincial governments, princely states; P. C. Ghosh made looms; shown at the 1901 Industrial Exhibition alongside the Congress session (Gandhi attended).
- **Swadeshi (1905)**, after Curzon's Partition of Bengal: debate — revive crafts, or build Western-style industry? Many Indians **rejected** Birdwood/Havell's anti-industrialism as a plan to keep India **"second-best" and dependent**; they wanted to **compete** with Britain. Textile mills, match factories, steamship lines started — many failed (little capital, weak technology).
- **Gandhi, *Hind Swaraj* (1909)**: reject "modern civilisation". Railways "spread plague, encouraged famine" — "beyond dispute that they propagate evil". Machinery is "the chief symbol of modern civilisation" and "a great sin"; Manchester destroyed Indian crafts; Bombay mill-workers are "slaves". Answer: go back to villages, **charkha and khadi** as the tools and symbols of **swaraj**. Little impact in 1909, but **huge impact after 1918–20**.

4.10 Chapter 4 in a nutshell

WHY THE BRITISH INTRODUCED THESE TECHNOLOGIES	WHY THEY HELD BACK / LIMITED THEM
• Military control and security (roads, railways, telegraph, survey) • Administration & communication over a huge land (mail, telegraph) • Trade & extraction (coal, cotton, export economy, British goods in) • Famine protection (railways, canals) — and revenue from irrigated land • Ideology: civilising mission, monuments, legitimacy after 1857	• Protect British industry (ships, locomotives, machinery, copper, iron) • Security: metals = arms (Arms Act 1878; Ajmer mines closed 1857) • Cost & unsuitable conditions (Indus steamers, steam ploughs) • Romantic anti-industrial view of India (Birdwood, Havell)

THE MAIN POINT ARNOLD WANTS US TO GET

1. Technology transfer was a **"dialogue", not a one-way diffusion** — and colonial power decided the terms: **"denial was often as important as diffusion"**.
2. The British chose technologies for **military, economic and ideological** ends. Most **benefits went to Britain**; India became **technologically dependent**.
3. **Indian environment, labour, skills and materials** reshaped every technology (bridge piers, anicuts, cables, ships); **old technologies survived** beside new ones (handlooms, wells, carts).
4. Technology carried **strong ideological meaning** — proof of British superiority — but Indians gave it their **own meanings**: use, enthusiasm, selective resistance, and finally critique (Swadeshi industry vs Gandhi's charkha).

CHAPTER 5

Imperial Science and the Indian Scientific Community

Book pages 129–168 · Period: 1858 (Crown rule) to the 1920s, with the peak of "imperial science" under Lord Curzon (1899–1905)

In one line: After 1858, science and the colonial state became close partners — **"imperial science"** used for better government and to **justify British rule** — but it was controlled by bureaucrats, racially exclusive and mostly "applied". At the same time, Indians built their **own scientific community** and turned science (especially **physics and chemistry**) into a source of **national pride**. The clash of these two worlds is the story of this chapter.

5.1 The big shift after 1858

- No sudden break from Company times, but a steady move from **exploring and observing** (with erratic Company interest) to a **confident alliance between science and the state**.
- The state used science for two things: **self-legitimation** (justifying its rule) and **more effective government**.
- But the state's view was **instrumentalist**: **"science existed to serve the empire, not to constitute an alternative source of authority or to dictate imperial priorities"**.
- From the 1890s to World War I, **unprecedented Indian interest** in Western science → an **Indian scientific community** that took part in international science.
- These two things — imperial science + Indian scientific community — pushed science to new prominence in the 1890s–1900s, but also created **tensions and contradictions**.

5.2 Science under the bureaucrats (the ICS)

- The **Indian Civil Service (ICS)** had little science training. It valued district experience, knowing villages and languages, and practical administration over "cosmopolitan" science. It kept the **scientific services under its control**. So state science was the science of the **"enlightened bureaucrat" — politically dependent**.
- **Census:** first systematic all-India census in **1871**, then every 10 years. More than numbers — a tool of **"governmentality"** that spread colonial ideas about India (religion, caste, health).
- **Gazetteers: W. W. Hunter** (ICS) proposed them in 1869 — "nothing [is] more costly to a Government than ignorance". *Imperial Gazetteer*: 9 volumes (1881), 14 volumes (1885–7), revised 1907–9.
  - Written for "practical administrators", not for scientists; run by an ICS officer, not a scientist.
  - The 1909 volume moves step by step: geology, weather, plants, animals → people (ethnology, language, religion) → government departments (public health, police). People's lives were "sandwiched between the zoology of fishes and the origins of the Aryan religion".
  - H. H. Risley's essay on caste begins with skull measurement (craniometry) as a "test of race".
  - After 1857, the gazetteers worked like a **"scientific title deed"** — proof to India and the world that Britain **knew** India, based on printed texts rather than "native" testimony.

SCIENTISTS VS BUREAUCRATS — CONSTANT TENSION

- **George Watt** (*Dictionary of the Economic Products of India*, 1889): publicly said it was for administrators; privately raged at **"Secretariat interference"** — "the curse of every Indian branch of special or scientific work".
- **Ronald Ross** resigned from the IMS (1899) in protest at poor research chances.
- **J. A. Voelcker** (agriculture report, 1893): a department of agriculture run by an ICS generalist (trained in classics, maths, literature) who moves on after a short time makes no sense — but even he accepted some civilian control was needed.
- Yet many British scientists **accepted** the system: India gave them unique opportunities and recognition.

Mixed results of state science

- **Fisheries:** Francis Day, Inspector-General (1870), criticised the salt tax — no effect; the department was **abolished** when he retired.
- **Botanical Survey of India** only in **1890** (George King) — mix of research and economic aims (cinchona, tea, coffee); small but effective; close links to **Kew**.
- **Meteorology:** after the Calcutta cyclone (1864) and Orissa famine (1866) → **Meteorological Department (1875)** under H. F. Blanford; 100+ observatories by 1878.
- **Astronomy:** N. R. Pogson (Madras) — 5,000 stars; his claim that **sunspots affect droughts and famines** (to the 1878 Famine Commission) got funding → **Kodaikanal Observatory (1895)**, famous for solar physics.
- **Famine pushed science** — but the first response was still railways, canals and the Famine Code (1880); **science was only secondary**.

5.3 "Imperial science" under Elgin and Curzon

- A 1884 proposal for a department of science was **rejected**.
- **Lord Elgin (Dec 1898)**, during plague unrest, admitted the government was **"almost without competent advisers in many branches of science"** → "misdirected action". He asked **Britain's Royal Society** to guide research in India → **Indian Advisory Committee of the Royal Society** (inactive till 1903).
- Arnold: this was a **backward step** — just when Indian scientists were emerging, the government showed no confidence in them and tied Indian science **even more to London**.
- **Curzon (1903)** set up a **Board of Scientific Advice** (heads of the scientific services, meeting twice a year) — but it had to report **every year to the Royal Society in London**. Curzon's reason: the Government of India owned "the largest landed estate in the world", agriculture was the backbone, resources were poorly explored, and research money was limited.
- Friction: London delays; only **"applied", not "fundamental"** research was thought suitable for India; Indian experts disliked London's "carping criticism". The link broke down by 1910; the London committee died after 1914, the Indian Board in 1922.
- **Roy MacLeod** sees here a "new sense of scientific independence". **Arnold disagrees**: real scientific independence (like political) came **only after 1945**; the story actually shows the **failure** of scientists in India to win autonomy from the state and London.

What science meant to Curzon

- Experts should **advise, not administer**. But science could be a force for change, a tool for **efficient government**, and — against rising nationalism — a **"fresh source of legitimation for British rule"**.

- **1899 speech to doctors:** the British came to India as **benefactors** bringing law, religion, literature and science. People may question British law and religion, but not science — especially **medicine**: "built on the bed-rock of pure, irrefutable science", "a boon … offered to all"; it lifts purdah "without irreverence" and breaks caste "without sacrilege". **Medicine alone was justification for British rule.**
- **Pusa (April 1905):** the central government must "give the lead" — model institutes, top (European) experts, results passed down to provinces and peasants. India as **one "scientific empire"**, not separate provinces.
- **The catch:** he asked princes and old elites to **fund** research, but was **unenthusiastic about Indian-proposed projects**. "The full credit for scientific and technological progress should rest with the imperial regime."
- **New institutes:** veterinary (Muktesar, 1898), agriculture (**Pusa, 1905**), forestry (Dehra Dun, 1906). By 1914, nearly a dozen **all-India ("imperial") services**: medicine, meteorology, veterinary, botany, agriculture, forests, geology.

> **TABLE 5.1 (1920): WHO STAFFED THE SCIENTIFIC SERVICES?**
>
> **195 Europeans vs only 18 Indians** across 11 services. Europeans were also paid far more — e.g. Agricultural Service Rs 1,000 vs Rs 460 a month; Educational Service Rs 1,910 vs Rs 490. Indians held **less than 10%** of scientific posts by 1920.

The "service ethos" — and its dark side: racism

- The services were small, **European and racially exclusive**, with an "esprit de corps" — science as **patriotic and paternalistic duty** to the empire. The 1916–18 Industrial Commission (Sir Thomas Holland) even called scientific workers a "caste".
- Deepak Kumar calls this **"apartheid"** in Indian science.
- **Pramatha Nath Bose** (London-trained geologist) — the first Indian in the Geological Survey. Its head **H. B. Medlicott** (1880): "he is a Bengali and may be physically unfit"; spoke of his "scientific helplessness"; said Indians were **"utterly incapable of any original work in natural science"**; "not expect them to run before they can walk". Bose resigned (1903) when Holland, **10 years his junior**, became Director. He joined the princely state of **Mayurbhanj** and found the **iron ore** that later fed **Tata's steel works at Jamshedpur**.
- In 1874, Bengalis were kept out of Geological Survey apprentice posts for "deficiency of stamina and manliness"; "manly" Sikhs were chosen instead. Valentine Ball: Indians were "deficient in synthetical powers".
- **Universities** were split into "two distinct racial camps". **P. C. Ray** (Edinburgh D.Sc.) was kept in the lower provincial service. **J. C. Bose** got into the Indian Educational Service but was paid **one-third** of British colleagues' pay until 1903.
- **Arnold's balance:** racism did hold Indians back (especially in medicine), but it also made some, like Ray, **more determined**. Many Indians were also inspired by European teachers (Ray by Alexander Pedler; Bose by Father Lafont). Bose even got a special research grant (Rs 2,000 a year, 1897) — though an official sneered he might be "an original thinker (a rare thing in an Indian)" and that salaries are "paid for definite services", not for "high intellect" — showing dislike of **research itself**.

5.4 Medical science: from neglect to "imperial" research

1880s–90s: India was behind

- **Ross**: no medical libraries; Pasteur's and Koch's discoveries "scarcely recognised, or … ridiculed".
- **Ernest Hart** (editor, *British Medical Journal*, 1894): why were all tropical disease discoveries made by the French, Germans, even Japanese — not the British? Because research was crushed by administration; a man with a new

scientific idea was seen as "**a sort of pestilential fellow with a new bug**"; promotion by seniority encouraged "respectable conservatism".

- Change came from **outsiders**: **E. H. Hankin** (1892) and **Waldemar Haffkine** (1893; anti-cholera, then anti-plague vaccine in Bombay, 1896) — neither was IMS. **Ross** proved mosquitoes spread malaria (Nobel 1902) — "against the odds"; a colleague told him "**We IMS men are not meant for research … we are simply doctors**".

Plague (1896) — the turning point

- Plague hit Bombay in 1896: 400,000+ deaths by 1900, ~8 million by 1914. Europe feared a new Black Death; at the **Venice sanitary conference (1897)** an **embargo on Indian shipping** was threatened.
- To save trade, the government passed the **Epidemic Diseases Act (1897)**: forced hospitalisation, destroying houses, examining rail travellers, banning fairs and pilgrimages → **widespread resistance**.
- **Malkowal (1902):** 19 villagers died of tetanus from a contaminated anti-plague vaccine → Haffkine blamed and dismissed (1904); cleared years later.
- 1898–99: government became **pragmatic** — eased coercion, accepted many deaths, and kept **political control over medical policy**. But the 1901 Plague Commission and British pressure made it take **research seriously**.

1900–1914: the research system

- Research kept in the **state, not universities**. Central Research Institute, **Kasauli (1906)**; Pasteur Institutes (anti-rabies, Kasauli 1900 and others); Bombay lab (later **Haffkine Institute**, 1925); **King Institute, Guindy** (1904); a Bacteriological Department (1906).
- **Indian Research Fund Association (IRFA), 1911** — funded research (Rs 5 lakhs a year; even before Britain's Medical Research Council); *Indian Journal of Medical Research* (1913); **Calcutta School of Tropical Medicine** (1921, Leonard Rogers).
- **Good:** India became known for tropical-disease research (malaria, cholera, kala-azar, plague).
- **Bad:** vaccine production swamped research (esp. in wars); institutes scattered; the **IMS re-took control** (Haffkine was called a "poor administrator"); research now became a reason to join the IMS (a Nobel like Ross!).

> **HILL-STATION SCIENCE: RACISM + ISOLATION**
>
> Research labs were put in cool **hill stations** — Kasauli (6,000 ft, near Simla), Muktesar (7,000 ft; moved from Poona partly because of public hostility to animal experiments). The belief: **serious science needs a cool climate and "temperate" races**. Viceroy **Lansdowne (1891)**: original research needs qualities not found in "races bred in a tropical climate". A Bombay principal claimed a frog's heart beats 48 hours after removal in Scotland but only 5 minutes in Bombay! Result: medical research became a **privileged, mostly white** pursuit, denied to under-funded universities. Isolation also bred boredom, rivalries and Indian criticism that science was **hidden from the public**.

5.5 Forestry and agriculture

Forestry: science + revenue

- **Richard Grove:** Company surgeons pioneered **conservation** — they linked deforestation to drying climate and famine; colonies were at the "cutting edge" of science.
- **Gadgil & Guha:** the real motive was **money** — Forest Department (1864), Forest Acts (1865, 1878) to tap forest wealth and get **timber for railways**. **Deepak Kumar:** foresters were "basically estate-managers"; revenue "was the only obsession".

- **Arnold:** both overstate. Forestry was **profitable** (a model for other departments) *and* genuinely scientific — it shows the **interdependence of science and state**. Huge domain (249,000 sq miles by 1916).
- **Dietrich Brandis** (German botanist), Inspector-General of Forests from **1864** for 19 years — brought German/French scientific forestry; founded the forestry school at **Dehra Dun (1878)** to train Indian assistants; wrote forest floras; used Indian plant names. For him, science and revenue were "largely, and legitimately, indistinguishable" — but forests should ultimately benefit the people.
- **Conflict:** restrictions on grazing, shifting cultivation, fuel → tribal revolts (1870s–80s) and **forest satyagrahas** (1920s–30s). Two knowledge systems clashed: professional forestry vs local customary knowledge.
- **But also engagement:** tribals worked as labour; their knowledge helped identify species and collect drugs (e.g. kino by Kurumbars in Malabar). Foresters slowly learned that "primitive" practices — shifting cultivation, grazing, burning grass — were needed to regenerate **sal and teak**.

Agriculture: neglected, then pushed by famine

- Before the 1890s, agriculture was **neglected** as a science — the state relied on big public works (canals, railways). Forestry got a service (1864) and school (1878) **30+ years before** agriculture. New crops focused on **commercial** ones (tea, cotton, cinchona), not peasant staples (rice, wheat, millet). Imported machines failed.
- **Famine** was the spur each time: Orissa 1866 → a department (closed 1879 to save money); 1876–78 famine → provincial departments (mostly data collection); agricultural chemist proposed in 1882, approved only in 1889. **Voelcker (1893)** praised irrigation but criticised neglect of soil chemistry and manures.
- The **1896–1900 famines** → 1901 Famine Commission: expert research is "the crying necessity of the time". Inspector-General of Agriculture (1901); **Indian Agricultural Service (1906)** — 42 years after forestry.
- **Pusa Institute** (Bihar, 1905–06) — funded by a £30,000 gift from American philanthropist **Henry Phipps**; 1,300-acre farm; "full staff of European specialists" + "native assistants". Of 40 posts, **only 8 Indians**, mostly junior.

> **ALBERT & GABRIELLE HOWARD — LEARNING FROM INDIAN FARMERS**
>
> Voelcker had said: don't import foreign seeds; test **indigenous varieties**. The Howards (Gabrielle was a rare woman scientist in colonial India) studied 3 dozen Indian wheats and bred **rust-resistant "Pusa" wheats**. They said trying to reform Indian farming on Western lines was a "**fundamental mistake**"; Indian practices were "**worthy of respect**"; the goal was "the application of Western scientific methods to the local conditions to improve Indian agriculture **on its own lines**". Pusa also studied how rice fixes nitrogen (with chemist Jatindra Nath Sen).

Pusa's problems: provinces resented its advice; isolation; Europeans needed big salaries, palatial bungalows, a Europeans-only club and "the finest tennis courts in India"; loss of purpose. After the 1934 Bihar earthquake it moved to Delhi — Arnold sees this as **symbolic of the fate of imperial science** and its secluded institutes.

5.6 The rise of an Indian scientific community

Who were they? Where did they come from?

- **Western education** in Calcutta (Hare School, Presidency College, St Xavier's) + study abroad (Gilchrist scholarships): **Mahendralal Sircar** (MD 1863), **P. N. Bose** (London, 1874), **J. C. Bose** (Cambridge & London), **P. C. Ray** (Edinburgh, 1882).
- **But also Indian roots**: most were bhadralok (Brahmin, Kayastha, Vaidya) with a heritage of learning and some wealth. J. C. Bose came from Vikrampur (an old centre of Sanskrit and Buddhist learning); his Brahmo father steered him away from the ICS. Ray's father was a Persian scholar; Ray joined the Brahmo Samaj. Sircar came from

a lower-ranking Sadgop family. **C. V. Raman**: Tamil Brahmin, mother from Sanskrit pandits; joined the Finance Department (1907), luckily posted to Calcutta.

- **Going abroad** gave a sense of belonging to an **international scientific community**. P. N. Bose dreamed of "a niche in the modern Temple of Science". Ray: "every civilised country including Japan was adding to the world's stock of knowledge but … unhappy India was lagging behind". Coming home to racism made them **more determined** — a "dual dedication to **science and nation**".

How small the scene was (Asiatic Society, 1895)

295 members, 62% non-Indian; many ICS, IMS and army. Over half the Indian members were princes, zamindars and notables. When J. C. Bose presented his famous paper on electric waves (May 1895), **only 13 members and 3 visitors** turned up. The state dominated; scope for Indian scientists was small.

Wider interest across India

- **Aligarh Scientific Society (1864)** — Sir Sayyid Ahmad Khan; translated ~40 works into the vernacular; *Akhbar Institute Gazette* (1866). **Bihar Scientific Society (1868)**. Urdu health journal *Guardian of Health* (Lahore, 1878–88) mixing "English and Native" ways.
- **Bengal** was most active: Society for Translating European Sciences (1828); Rajendralal Mitra's science journal; between 1875 and 1896, **776 science publications in Indian languages** in Bengal alone.

INDIAN ASSOCIATION FOR THE CULTIVATION OF SCIENCE (IACS), 1876 — MAHENDRALAL SIRCAR

- Sircar (an MD who later turned to homoeopathy) saw science as the source of the **"superiority of the Western nations"** and the key to India's progress and self-respect.
- In 1869 he urged Indians to shake off "indolence and apathy", be **self-reliant**, and not look to the state; he wanted an institution "**entirely under native management and control**", with public lectures for ordinary people. It took 7 years to raise the money.
- It proved that **Indians were fit for "pure" science and research**, not just artisans or "departmental drudges".
- It turned Indian science towards **physics and chemistry** — sciences that did not put India and Indians "under scrutiny" (unlike colonial natural history and medicine) and that helped Indians **connect with their own past**.
- Sircar: science has "a **higher and nobler claim than the narrow, utilitarian, Benthamite one**" — it ennobles the mind. He died disappointed (1904) at public apathy, but IACS inspired **J. C. Bose, P. C. Ray and C. V. Raman**, and made **Calcutta the centre of national science**.

Other forces that helped

- **Nationalism and Swadeshi (1905)** — emphasised **practical** science and technical education: P. N. Bose's pamphlet on technical education (1886) and Indian Industrial Association (1891); Association for the Advancement of Scientific and Industrial Education (1904) to train Indians abroad; Dawn Society (1904) → National Council of Education → Bengal Technical Institute → later **Jadavpur** College of Engineering and Technology; and the University College of Science, Calcutta.
- **Universities:** before 1900, little research; chairs reserved for IMS/Educational Service men. Curzon's Universities Commission and the **Indian Universities Act** allowed postgraduate teaching and research. From **1909**, **Asutosh Mukherjee** (Vice-Chancellor, Calcutta) started postgraduate science; chairs in physics, chemistry, maths, botany, zoology.
- **Indian philanthropy** filled the gap left by the state: **Tarakanath Palit and Rash Behari Ghosh** gave Rs 2.4 million for physics and chemistry chairs (the Palit chairs reserved for Indians); princely states (Baroda, Hyderabad, Mysore, Travancore) funded technical education.

- **J. N. Tata (1896)** — Rs 3 million of property for an Indian university of science. **Curzon narrowed it**; Bangalore chosen partly because Mysore paid and the **climate suited European staff**. The **Indian Institute of Science** opened (1908–09) but struggled; ran under Europeans until **C. V. Raman became its first Indian director (1933)**.

> **INDIAN SCIENCE CONGRESS (1914) — FROM COLONIAL TO NATIONAL SCIENCE**
>
> First met at the Asiatic Society; President **Asutosh Mukherjee** praised Jones, but urged landlords and businessmen to see science as key to agriculture and "industrial regeneration", and said even "the most enlightened governments" must be reminded of science's claims on public funds. Ironically, the idea came from two **Europeans** (MacMahon and Simonsen), who wanted something like the British Association. One-third of the 35 papers in 1914 were by Indians. When **P. C. Ray** became President (1920, a strongly nationalist address) and **Raman** secretary (1921), the Congress became the **forum of national science** — as the Asiatic Society had been for colonial science.

5.7 Chemistry, physics and plant physiology: the Indian breakthrough

Why physics and chemistry?

- Chemistry had been neglected in Company times (P. N. Bose's 1883 survey found hardly any chemistry papers). Chairs went to IMS men by seniority (O'Shaughnessy is remembered for the telegraph, not chemistry).
- **No British dominance and no state services** in physics and chemistry → **open to Indians** (unlike medicine, where Europeans held tight control).
- Their **theoretical** nature suited Indian intellectual traditions better than botany and geology — the utilitarian field sciences of "exploration and exploitation".
- The British woke up late: German **synthetic indigo** ("blue terror", Curzon) ruined indigo planters; chemistry became vital for agriculture and industry. The Industrial Commission (1918) proposed an all-India Chemical Service.

P. C. RAY — CHEMISTRY AS A "SUBVERSIVE SCIENCE"

- Paid only Rs 250/month as Assistant Professor at Presidency College despite an Edinburgh D.Sc.
- Discovered **mercurous nitrite** (1895) — reprinted in *Nature*.
- Founded **Bengal Chemical and Pharmaceutical Works (1899)** — 1,400 workers by 1926; proved Bengalis could run big business and link lab to industry.
- Built a **"School of Indian Chemistry"** — 100+ papers by 1920; students placed across India; a guru–disciple (*acharya*, ashram-like) bond.
- **Opposed the Chemical Service (1920)**: the services were "an anomaly, nay, a glaring anachronism"; Indian graduates could "hold their own" — fund universities instead.
- Chemistry was "subversive": the kind of **pure science Indians were supposed to be unfit for**, not a British monopoly; inspired by ancient Indian alchemy, French and German chemists, and Japan.
- 1920: "**The fate of a nation will henceforth depend more upon the achievements of its students of science than upon the skill of its generals**."

J. C. BOSE AND C. V. RAMAN — PHYSICS

- **J. C. Bose** — reluctantly admitted to the Educational Service; researched **electric (radio) waves** extending Hertz, with simple home-made equipment; practical use was later taken by others like Marconi. Praised in London (1896), Paris, Berlin — "the first Indian to win distinction through investigation in science" (Geddes).
- Kelvin and Lister asked for a proper lab for him — the Bengal government refused. Bose: "**the real laboratory is one's own mind**".
- Bose and Ray "invented national science for India as **laboratory science**", unlike the **field sciences** dominated by Europeans.
- Later Bose studied **responses in plants and metals** (making plants "speak"). British physiologists mocked it as "Oriental imagination" — maybe cost him a Nobel. (Raman called it "mumbo-jumbo"!)
- **C. V. Raman** built on Bose's path — cheap "sealing wax and string" physics — and won the **Nobel Prize (1930)** for the **Raman effect**.
- Birbal Sahni on Bose (1937): Indians were "laughed at" as scientists; Bose was "**the pioneer who broke this spell**".

5.8 Chapter 5 in a nutshell

WHAT "IMPERIAL SCIENCE" WANTED	ITS LIMITS AND CONTRADICTIONS
• Efficient government (census, gazetteers, services) • Legitimacy against nationalism ("medicine alone justifies British rule") • Answers to famine & epidemics (agriculture, meteorology, plague research) • Revenue (forestry, economic products) • One centralised "scientific empire" led by top European experts	• Science kept under ICS bureaucrats & London's Royal Society • Mostly "applied", not fundamental research • Racially exclusive (195 vs 18 in 1920; Medlicott vs P. N. Bose) • Isolated hill-station institutes, cut off from people and universities • Credit had to go to the Raj; Indian projects discouraged

THE MAIN POINT ARNOLD WANTS US TO GET

1. After 1858 science became a **partner of the colonial state** — for governing and for **legitimising** British rule (peak: Curzon). But it was **instrumental, bureaucratic, racially exclusive and dependent on London**.
2. At the same time, an **Indian scientific community** arose — through Western education *and* Indian intellectual heritage, nationalism, universities and Indian philanthropy.
3. Indians deliberately chose **laboratory sciences (physics, chemistry)** to prove they could do **original, "pure" science** — turning science into **national pride** and a challenge to colonial racism.
4. The two worlds — **state services vs universities**, **imperial science vs national science** — were "competing but deeply entangled". This tension shaped science in India up to independence.

EXAM SECTION

Essay Plans: Past Questions + Likely Questions

Each plan gives you: what the question is really asking → a **thesis line** (your main argument) → an **intro** → **body points with examples** → a **conclusion**. For a 1000-word answer, write an intro (~120 words), 6–8 body paragraphs (~100 words each), and a conclusion (~100 words).

THE WINNING FORMULA FOR ANY ESSAY IN THIS COURSE

1. **Intro:** define the key words of the question + say Arnold's big idea (science is social/political; interaction, not one-way diffusion; but under unequal colonial power).
2. **Body:** one point per paragraph. Every point = **claim + concrete example (name, date, number) + why it matters**.
3. **Show both sides:** British motives AND Indian agency; what worked AND its limits; historians who disagree (e.g. Stone vs Whitcombe, Desmond vs Arnold, MacLeod vs Arnold).
4. **Name-drop historians** correctly: Basalla, Adas, Bayly, Edney, Headrick, Pacey, Kerr, Grove, Kumar, Sangwan, Prakash, Chatterjee. It shows you read.
5. **Conclusion:** come back to Arnold's main message and answer the question directly in one clear sentence.

A. The 6 past test questions

TEST 1 · Q1

Discuss the various characteristic features of 'modern' science.

Main source: Ch 1 (+ examples from Ch 2, 5). This question may also draw on your class lectures on the Scientific Revolution — add those points if you covered them.

THESIS

Modern science is usually described as empirical, rational, systematic and universal. But as Arnold shows, it is also a **social institution** — organised, professional, tied to the state, industry and power. In colonial India, these features became very visible, and "being scientific" became the main sign of being "modern".

BODY POINTS

1. **Observation, measurement and experiment over authority of texts** — "the measuring-rod, the theodolite and the telescope" vs Puranic astronomy (1844); Lambton's chains and Everest's compensation bars; Bose's sensitive lab instruments.
2. **Systematic classification and a standard language** — Linnaeus's binomial system (1753); Indian plant names replaced by European taxonomy; "testimony of nature" over "testimony of natives".
3. **Claims to universality and objectivity** — science presented as "value neutral, objective… universally valid" (Basalla, as criticised by Adas). Arnold: actually it is "a highly social activity", culturally specific.
4. **Organised in institutions and networks** — Royal Society (1662), Asiatic Society (1784), journals (*Asiatic Researches*, *Nature*), museums, botanic gardens; recognition through medals, FRS, Nobel Prize; "invisible colleges".
5. **Specialisation and professionalisation** — from "gentleman amateurs" (army surgeons, engineers) to professional bodies (Oldham & GSI 1851/56; scientific services by 1914; IRFA 1911).

6. **Utility: linked to state, economy and industry** — "useful knowledge", "improvement"; coal for steam, cash crops, chemistry for industry (synthetic indigo); science + technology fused in the steam age.
7. **Linked to power** — knowledge to rule: surveys, census (1871), gazetteers ("scientific title deed"), "governmentality".
8. **International and cumulative** — a world community adding to "the world's stock of knowledge" (P. C. Ray); Indians going abroad to join "the modern Temple of Science" (P. N. Bose).
9. **A rational, secular world-view and the idea of progress** — modernity defined by "advanced technology and the spirit of science, … a rational view of life, a secular approach" (Pye); "scientific reasoning became the organising metaphor" of the Indian elite (Prakash).
10. **Laboratory vs field science** — Bose, Ray and Raman made physics and chemistry (lab sciences) the face of Indian modern science.

CONCLUSION

Modern science is not only a method; it is an institution with authority. In India it came with colonialism, so its "modern" features also acted as **markers of European superiority** — which Indians then claimed for themselves.

TEST 1 · Q2

Discuss the nature of the relationship between science and the East India Company in its early years in India.

Main source: Ch 2

THESIS

The relationship was **erratic, opportunistic and utilitarian** — a "fitful flirtation" rather than "rule by science". The Company supported science when it served **profit, war or prestige**; otherwise science depended on individual army engineers and surgeons working in their spare time.

INTRO IDEA

The EIC (1600) was "as old as modern science itself" — its rise in India ran parallel to Linnaeus (1753), Plassey (1757), Lyell and Darwin. It was well placed to use science, but how central was science to it?

BODY POINTS

1. **Gatekeeper and monopolist** — Court of Directors controlled entry (Humboldt refused; Hooker an exception), appointments and funding; London scientists (Banks, Lyell, Darwin) depended on Company servants.
2. **Stingy, erratic patron** — Roxburgh's book a "commercial disaster" (£300 → £2,000); only 40 guineas for the Siwalik catalogue vs £1,000 from government; "cheese-monger Emperors".
3. **Self-interest + prestige** — Desmond says not purely commercial (library, museum); Arnold: being "munificent patrons of science" was politically useful; Wellesley used surveys to boost his image (Vicziany).
4. **Utilitarian surveys = "governmentality"** — Mackenzie (Mysore 1799–1810), Buchanan (1800–1, 1807–14): resources, revenue, people; plants "useful or rare or curious". Hove's cotton spying (1787).
5. **Mostly "gentlemanly" spare-time science** — officers busy with duties; few societies; Asiatic Society (1784) mainly literary, only 119 members in 1845; journals folded ("it would not pay").
6. **Army engineers and surgeons as all-purpose experts** — 28 of 42 botanists were surgeons; professionals only late (Oldham 1851: "an office, a clerk and a chaprassi").
7. **Where the Company did commit: utility** — Great Trigonometrical Survey (1818) for war and control (India's first dedicated scientific service); Calcutta Botanic Garden (1787) for profit and famine; Coal Committee (1836) for steam.

8. **Dependent and isolated science** — Lyell's condescension; Oldham: "a tradesman without his tools"; data sent to Europe for theory.
9. **Science "about India, not for Indians"** — no Indian members of the Asiatic Society till 1829; but Indians still engaged (Bal Shastri Jambhekar; Dwarkanath Tagore).
10. **From Orientalist respect to dismissal** — astronomy and plant names; India framed as "tropical" and in need of rescue.

CONCLUSION

"The British enjoyed the company of science; it would be excessive to suggest that they ruled by it" (p. 25). But by the 1830s–50s, practical needs (maps, coal, steam) were pushing the Company towards professional state science — the base for the Crown's "imperial science".

TEST 2 · Q1

Science and technology were loaded with diverse functions and meanings in colonial India. Discuss.

Main sources: all four chapters — this is the best "show everything you know" question.

THESIS

In colonial India, science and technology were never just about knowing nature. They worked as a **tool** (profit, war, control), a **symbol** (superiority, civilising mission), a source of **legitimacy**, a marker of **identity** — and later as a **weapon of Indian nationalism** and a **target of critique**. Their meaning depended on who used them and why.

BODY POINTS (PICK 7–8)

1. **Economic** — Kyd's garden to "outstrip our rivals" (cinnamon, indigo, tea); coal; forestry revenue; railways to "intermesh" Britain and India's economies.
2. **Military/strategic** — surveys after Tipu (1799); Grand Trunk Road; Dalhousie's 1853 railway minute; telegraph in 1857; Arms Act (1878) and closing of Ajmer mines.
3. **Knowledge for control** — Buchanan's surveys, GTS as "panopticon" (Edney), census 1871, Imperial Gazetteer as "scientific title deed".
4. **Civilising mission and superiority** — Hardinge: steam "the greatest instrument of civilisation"; Dalhousie: "great engines of social improvement"; Macaulay mocking Indian astronomy; Edney: "they did science, the Indians did not".
5. **Legitimacy after 1857** — technology as safer than social reform; Queen's Proclamation; Curzon: medicine "alone" justifies British rule; Begum of Awadh's sarcastic reply.
6. **Identity and prestige of Europeans** — science as a badge of the elite (Edney); medals and FRS as career springboards (Falconer); Company as "munificent patron".
7. **Humanitarian face** — famine: botanic gardens, "protective" railways (1880), canals, agricultural research (Pusa).
8. **Religious, aesthetic and emotional** — Carey's botany praising God; Himalayas as a "spiritual homecoming"; Cotton's belief that irrigation would spread Christianity.
9. **Racial meaning** — tropicality; hill-station labs; Lansdowne on "races bred in a tropical climate"; Medlicott on Bengalis.
10. **For Indians: modernity, reform, nationhood** — Prakash's "organising metaphor"; Sircar's IACS (science = source of Western superiority); Ray: "the fate of a nation…"; Bose "broke this spell"; Swadeshi industry.

11. **Critique** — Keshab Sen ("materialistic"); Birdwood/Havell's romantic anti-industrialism; Gandhi: machinery "a great sin", railways "propagate evil"; charkha as counter-symbol.
12. **Everyday meanings** — railways for pilgrims and travel (*Pather Panchali*); telegrams for business and politics; rumours and resistance to plague measures.

CONCLUSION

Science and technology were "multi-meaning": the same railway was a tool of empire, a symbol of civilisation, a carrier of pilgrims and a sign of slavery for Gandhi. This proves Arnold's point that science is **social and political**, never neutral.

TEST 2 · Q2

Analyse the multiple ways in which the 'centre' and 'periphery' influenced each other (with regard to science and technology) during East India Company rule.

Main sources: Ch 1 (Basalla & critics) + Ch 2 (examples). Stay within c. 1757–1858.

DEFINE FIRST

Centre = the metropole: Britain/London — Court of Directors, Royal Society, Kew, British scientists, British industry. **Periphery** = the colony: India — Company servants, Indian informants and workers, Calcutta/Bombay/Madras institutions.

THESIS

Basalla's diffusion model sees a **one-way flow** from centre to periphery. Arnold shows influence ran **both ways** — India supplied data, discoveries, techniques and careers that changed metropolitan science — but always inside an **unequal, political hierarchy** where the centre kept authority.

CENTRE → PERIPHERY

1. **Control and patronage** — Court of Directors controlled access, money, jobs; refusal to Humboldt.
2. **Institutional models** — Royal Society → Asiatic Society (1784); BAAS as the model (M'Clelland's failed attempt).
3. **Methods and instruments** — Linnaean taxonomy (Koenig 1778); triangulation from the British Trigonometrical Survey and Irish Ordnance Survey; Colby's bars; Oldham from Ireland.
4. **Standards and judgement** — Indian astronomy judged by European standards (Trautmann); Indian plant names replaced; Lyell's condescension ("remote confines").
5. **Economic demands** — Banks's vision of India as a tropical estate; Board of Trade's spy Hove; London lobby against Indian copper/mineral competition.

PERIPHERY → CENTRE

1. **Raw data and specimens** — 40 tons of Siwalik fossils; plants to Kew; Darwin's India informants (Falconer, Royle, Blyth).
2. **New knowledge that changed metropolitan science** — Siwalik fossils for debates on extinction and climate; Deccan traps, laterite; Royle's tropical–temperate–alpine zones; the idea of "tropicality".
3. **Innovation ahead of Europe** — GTS "overshadowed all European geodetic operations" (Cannon), measured the earth's shape; Ganges Canal became a model for the USA and Australia; Bombay-built ships for the Royal Navy.
4. **Colonial careers became metropolitan careers** — Falconer (Wollaston Medal 1837, FRS 1845), Royle (King's College), Everest, Hooker.
5. **Indian knowledge entered Western science** — the raja and the "tooth of Deo"; Royle on the antiquity of Hindu medicine; Jones on Sanskrit–Greek–Latin; Sikdar and the pandits; Indian artists' plant drawings.

6. **Conservation ideas** — Company surgeons' warnings about deforestation and climate (Grove argues the colonies were at the "cutting edge").

THE LIMITS (SHOW ARNOLD'S BALANCE)

- Isolation and dependence: Oldham's "tradesman without his tools"; "collecting and recording" for Europe to theorise; Falconer hating his return; Blyth stuck in Calcutta.
- Sangwan/MacLeod argue colonial scientists were equal and the metropolis "moves" — Arnold replies this ignores the "political character of science": a clear hierarchy put Britain on top. Indians stayed subordinate (Edney: "they did science, the Indians did not").
- Over time, conformity with the centre grew (plant names).

CONCLUSION

Centre and periphery shaped each other — it was an **interaction, not a diffusion** — but the interaction was **unequal**. As Arnold says, Basalla "trivialises the compulsions of colonialism"; yet ignoring the centre–periphery hierarchy altogether would be just as wrong.

TEST 3 · Q1

Highlight and discuss some of the salient features that are common to the introduction of almost all technologies during the era of British rule in India.

Main source: Ch 4 (+ Ch 2 for surveys, Ch 5 for institutes). Use railways, telegraph, irrigation, steamships, textiles, mining as your examples.

THESIS

The introduction of technology was not a simple "transfer" but a **"dialogue" on colonial terms**. Almost every technology shows the same pattern: **driven by state needs**, **supplied from Britain**, **restricted when it threatened British interests**, **reshaped by Indian conditions and labour**, **coexisting with older technologies**, **loaded with ideology**, and **received in mixed ways** by Indians.

BODY POINTS (COMMON FEATURES)

1. **State-driven for military, administrative and economic needs** — roads (GTR), railways (Dalhousie 1853: news and troops), telegraph (Rangoon 1852, 1857), Ganges steamers (mail), Calcutta Mint (1831).
2. **Imported from Britain → dependence; British firms profit** — 90% of textile machinery from Lancashire; prefabricated steamships; rails, engines, even drivers; 14,000 locomotives sold vs ~700 made in India; workshops only for repair.
3. **Denial as important as diffusion** — Indian shipbuilding blocked (Sangwan); locomotive making thwarted; Arms Act 1878; Ajmer mines closed 1857; no tariffs for handlooms.
4. **Adapted to Indian conditions ("in situ" innovation)** — O'Shaughnessy's thick insulated cables; Indian well-sinking for bridge piers; Ganges Canal "shaped bit by bit to the circumstances of the country"; took decades (railway building "routinised" only by the 1890s); some failures (Indus steamers, steam ploughs).
5. **Built on cheap Indian labour, tools and methods** — Kerr: labour "worth many a machine"; basket labour; Bhor Ghat 42,000 workers; earth-and-rubble anicuts.
6. **Indians kept in subordinate roles** — exceptions prove the rule: Shib Chandra Nandy (before posts were "Europeanised"), Cursetjee, Sikdar.
7. **Old and new side by side** — handlooms (1/3 of output in the 1930s), wells and tanks (~60% of irrigation in 1901), bullock-carts feeding railways.
8. **Famine as a driving force** — "protective" railways (1880 Commission: 20,000 miles); canals after famines (Godavari after 1833; Ganges Canal after 1837–38); roads by famine labour.

9. **Heavy ideology** — technology as proof of civilisation and superiority; monuments "like Rome" (MacGeorge, Deakin); used for legitimacy after 1857.
10. **Pushed by individuals and army engineers** — Bentinck, Dalhousie ("technophile"), Curzon; Cotton, Cautley; "technocratic paternalism".
11. **Unintended costs** — de-industrialisation of crafts; waterlogging, *reh*, malaria (Whitcombe vs Stone); disease spread by railways.
12. **Mixed Indian responses** — fast adoption (railway "naturalised in the native mind", 1858; pilgrims; telegrams), selective resistance (1857 strategic attacks; plague measures), critique (Keshab Sen, Gandhi) and Swadeshi counter-industry.

CONCLUSION

These shared features show Arnold's argument: colonial technology was at once a **tool of empire, a dialogue with India, and an act of denial** — and its meaning stayed contested right up to Gandhi's charkha.

TEST 3 · Q2

Discuss the place and roles of Indians and pre-existing Indian methods and materials during the introduction of technologies into India by the British colonial power.

Main source: Ch 4 (+ Ch 2 survey/fossils/botany, Ch 5 Howards/forestry)

THESIS

Against the "pure transfer" story, Indians and Indian methods were **central** — as labour, skilled artisans, intermediaries, technical assistants, entrepreneurs, sources of materials and techniques, and as users and critics. But colonial power kept them **subordinate**, often **looked down on** their methods, and sometimes **suppressed** Indian technology.

BODY POINTS

1. **Indian materials** — teak for ships (even steamer hulls like the *Hugh Lindsay*); kankar as "Indian macadam" on the Grand Trunk Road; Raniganj coal; Salem ore for Porto Novo steel; coir and lime in shipbuilding.
2. **Indian techniques adopted by the British** — rabbeting (no caulking); the well-sinking method for bridge piers; Grand Anicut's earth-and-rubble design copied by Cotton, who credited the "old native engineers"; restoring Mughal canals (Shah Jahan's Delhi canal).
3. **Indian labour as the backbone** — 42,000 on the Bhor Ghat; Santal and Bauri families in coal mines; basket labour on the Ganges Canal; customary tools beat wheelbarrows (Kerr); quick learning of new skills (riveting).
4. **Skilled artisans and intermediaries** — Parsi Master Builder Lowji Nusserwanji Wadia (159 ships at Bombay 1736–1821); Wudder well-diggers; jobbers in mills; Julaha weavers.
5. **Technical assistants and pioneers** — Shib Chandra Nandy (telegraph lines), Radhanath Sikdar and the pandits (survey), Ardaseer Cursetjee (engineer, FRS), Lalla Tantimul (horse *dak*, 1849).
6. **Indian entrepreneurs and capital** — Dwarkanath Tagore (Raniganj, 1836); Parsis → Tata; P. N. Bose finding Mayurbhanj iron ore for Tata; Swadeshi ventures; P. C. Ray's Bengal Chemicals.
7. **Pre-existing systems that continued and adapted** — handlooms with fly-shuttle, jacquard and electricity; wells and tanks; bullock-carts and country boats; Birdwood's praise of the "Mahratta plough"; the Howards' respect for Indian farming "on its own lines".
8. **Indians as users who gave new meanings** — railways for pilgrimage and self-discovery; telegrams for business and nationalism.

9. **Indian resistance and critique that shaped outcomes** — Banjaras and boatmen refusing to carry coal (Narmada); strategic attacks in 1857; resistance to plague measures; Gandhi's charkha.

BUT — THE LIMITS

- Indians strictly in **subordinate posts**; posts became "Europeanised" (telegraph).
- Indian methods **belittled** — "crude" tools (Buchanan), "art but no science" (Ball), crafts sent to the ethnographic museum or linked to "criminal tribes".
- Indian capacity **suppressed** — shipbuilding, iron, metals, locomotives ("denial as important as diffusion").

CONCLUSION

Colonial technology "nestled" with Indian technology in a **dialogue** (Pacey). Indian agency was **indispensable but constrained** — exactly Arnold's picture of interaction under unequal power.

B. Likely new questions (shorter plans)

LIKELY Q7 · CH 1

Critically assess George Basalla's model of the spread of Western science with reference to colonial India.

- **Explain** the 3 phases (data source → dependent colonial science → national science).
- **What fits India:** botany and zoology dominated (Kyd, Roxburgh, Buchanan); colonial scientists dependent on London (Royal Society; Lyell; Board of Scientific Advice reporting to London); Indian Science Congress (1914), IISc, Raman's Nobel look like a move toward "national science".
- **Critique (7 points)** — ignores medicine/technology; one model for all colonies; colonial science = Europeans only; science as neutral and its spread inevitable (Adas); colonial states blocked science ("denial", racism); Eurocentric/passive recipients (India's own traditions — Jai Singh, Ayurveda–Unani, wootz); ex-colonies stay dependent.
- **Evidence of interaction:** local knowledge (raja & fossils, Sikdar, Wudders, well-sinking piers); recipient cultures "subvert, contaminate, reorganise" (Raina & Habib); India sometimes ahead (GTS).
- **Arnold's middle way:** reject diffusion but keep "colonial science" as "technologies of power operating within and through science in a colonial setting".

LIKELY Q8 · CH 5

What was "imperial science"? Discuss its aims, features and limitations.

- **Definition:** post-1858, esp. under Curzon — India-wide, state-run, centrally directed science serving India and the empire, staffed by European experts (services, institutes).
- **Aims:** efficient government; legitimacy against nationalism (Curzon on medicine); famine and epidemics; revenue; India as one "scientific empire".
- **Features:** census, gazetteers; Board of Scientific Advice (1903); institutes (Muktesar 1898, Pusa 1905, Dehra Dun 1906, Kasauli 1906); IRFA 1911; nearly a dozen all-India services by 1914; "service ethos".
- **Limitations:** controlled by ICS ("Secretariat interference"); dependent on the Royal Society; mostly "applied"; racist (195 vs 18; Medlicott); isolated hill stations (Lansdowne; frog's heart); vaccine production swamped research; Pusa's isolation and move to Delhi; credit monopolised by the Raj (Tata's institute narrowed).
- **Conclusion:** it raised science's status but kept it subordinate, exclusive and dependent — provoking Indian "national science".

LIKELY Q9 · CH 5

Trace the emergence of an Indian scientific community between the 1870s and 1920s. What factors helped and hindered it?

- **Helped:** Western education (Hare, Presidency, St Xavier's) + study abroad (Gilchrist scholarships); bhadralok heritage of learning; IACS (1876) and Sircar's vision; vernacular science societies (Aligarh 1864, Bihar 1868, Bengal's 776 publications); Swadeshi and technical education (Dawn Society, Jadavpur); Universities Act and Asutosh Mukherjee (1909); Indian philanthropy (Palit, Ghosh, Tata, princes); Indian Science Congress (1914 → Ray 1920, Raman 1921).
- **Hindered:** racism in services (Medlicott vs P. N. Bose; pay gaps for J. C. Bose, Ray); small, state-dominated scene (13 people at Bose's 1895 talk); state kept research in services, not universities; Curzon's coolness to Indian projects; Western scepticism (Bose's plant work as "Oriental imagination").
- **Choices:** physics and chemistry (no British monopoly; theoretical; suited Indian legacies) → laboratory science as "national science"; Ray's "subversive science"; Bose "broke this spell"; Raman's Nobel (1930).

LIKELY Q10 · CH 1, 4, 5

How did Indians respond to Western science and technology? / Science and modernity in colonial India.

- **Enthusiasm:** Western-educated elite — science as "organising metaphor" (Prakash); Jambhekar's Marathi translations; Calcutta and Bombay welcomed railways; Keshab Sen praised steam.
- **Science for nationhood:** Sircar (IACS), Ray ("fate of a nation"), Bose, Raman, Nehru's "shrine of science"; Swadeshi industries; Tata.
- **Translation / hybridity:** science renegotiated in Indian idioms; selective use of Hindu ideas; Ray drawing on ancient Indian alchemy and metallurgy; Bose's plant responses; Urdu journals mixing "English and Native" medicine.
- **Critique:** Chatterjee's dilemma (nationalism accepting colonial premises of modernity); Keshab Sen's warning about materialism; Gandhi's *Hind Swaraj* and the charkha.
- **Popular responses:** pilgrims on trains; resistance to plague measures and survey parties; rumours — but British exaggerated "hostility" (1857 destruction was strategic).

LIKELY Q11 · CH 2, 4, 5 (GREAT CROSS-CHAPTER QUESTION)

Discuss the role of famine in shaping science and technology in colonial India.

- **Company period:** Bengal famine 1770 (~10 million) → Kyd's botanic garden (1787) for famine food plants; Roxburgh at Samalkot; India seen as "tropical" and vulnerable; surgeons link deforestation to drought (Grove).
- **Steam age:** canals after famines (Godavari after 1833; Ganges Canal after 1837–38); 1880 Famine Commission → 20,000 miles of "protective" railways; famine labour building roads; railways as *laissez-faire*-friendly famine policy.
- **Crown:** Orissa famine (1866) → meteorology and an agriculture department; Pogson's sunspot–famine claim → Kodaikanal (1895); 1896–1900 famines → 1901 Famine Commission → Inspector-General of Agriculture, Pusa, Agricultural Service (1906).
- **Limits:** technology favoured over science; canals' environmental costs; Gandhi: railways "encouraged famine".
- **Conclusion:** famine pushed science and technology towards **agrarian** rather than **industrial** goals — a very "Indian" pattern.

LIKELY Q12 · CH 1, 2, 4, 5

Discuss the relationship between Western science and indigenous knowledge in colonial India.

- **Pre-colonial plurality** — not a blank slate (Ch 1); Orientalist "triptych" vs revisionists.
- **Early dependence and respect** — Bayly's "information order" on Indian foundations; Jones's Sanskrit names; Royle on Hindu medicine; Jai Singh's observatories; raja and fossils; local guides for Hooker.
- **Growing dismissal** — Trautmann (European standard); Macaulay; plant names replaced; "testimony of nature over natives"; crafts as "crude" and to the museum; Edney's "they did science, the Indians did not".
- **Practical borrowing continued** — well-sinking, anicuts, rabbeting, Wudders; foresters learning from shifting cultivators; tribals' knowledge of species and drugs.
- **Late re-appreciation** — Howards and Voelcker (indigenous wheat); Birdwood/Havell on crafts; Brandis on Indian plant names.
- **Indian re-use of tradition** — Ray drawing on ancient alchemy; Sircar's physics/chemistry connecting Indians to their past; Gandhi's charkha.

LIKELY Q13 · CH 5

Examine the relationship between science and the colonial state after 1858.

- **Alliance:** science for legitimacy and governance (census, gazetteers, services, institutes, Curzon).
- **Subordination:** "science existed to serve the empire"; ICS control; Watt's "Secretariat interference"; Ross resigns; Day's fisheries abolished; Voelcker on generalist directors.
- **Dependence on London:** Royal Society Advisory Committee; Board of Scientific Advice reports to London; only applied research (MacLeod vs Arnold).
- **Exclusion:** racial "apartheid" (Kumar); Table 5.1; hill stations.
- **State vs universities:** research kept in state institutes; Ray attacks services as "a glaring anachronism"; Indian philanthropy funds universities.
- **Conclusion:** a "competing but deeply entangled" relationship — the state made science big but not free.

LIKELY Q14 · CH 4

"Modern technology was the British Raj's greatest monument." Discuss with reference to railways and irrigation.

- **British view:** Hardinge, Dalhousie, MacGeorge, Deakin ("would fully justify their supremacy"); Ganges Canal unsurpassed; 40,000 miles of track; technology over social reform after 1857.
- **Real gains:** faster transport and news; famine relief; irrigated 1 million acres; Stone's optimistic view; railway workshops as "schools of skills".
- **But:** "a British show" — imports, 14,000 vs 700 locomotives; built for war and trade; dependent on Indian labour and methods; environmental costs (Whitcombe); old wells (60%) and carts still essential; Gandhi's critique; the Begum of Awadh's jibe.
- **Conclusion:** a monument — but to colonial priorities, not to balanced Indian development.

LIKELY Q15 · CH 4

Discuss the debate on industrialisation and technology in late colonial India (Birdwood, Havell, Swadeshi, Gandhi).

- **Western romantic critique:** Ruskin, Morris → Birdwood (crafts as art; steam-plough worshipped as a lingam); Havell (revive handlooms with the fly-shuttle; avoid European industrial evils).
- **Indian suspicion:** anti-industrialism seen as keeping India "second-best" and dependent; Swadeshi (1905) mills, match factories, steamship lines — many failed.
- **Indian modernisers:** P. N. Bose on technical education; Ray's Bengal Chemicals; Tata; Science Congress urging industrial regeneration.
- **Gandhi:** *Hind Swaraj* (1909) — machinery "a great sin", railways "propagate evil", charkha and khadi as swaraj; huge influence after 1918–20.
- **Link to Ch 1:** Nehru vs Gandhi; post-1947 compromise favouring modern science.

LIKELY Q16 · CH 5

Medicine as "imperial science": discuss with reference to plague and medical research.

- **Neglect before 1890s** — Ross, Hart ("pestilential fellow with a new bug"), "we are simply doctors".
- **Outsiders lead** — Hankin, Haffkine; Ross's Nobel (1902).
- **Plague 1896** — international pressure (Venice 1897); Epidemic Diseases Act; resistance; Malkowal (1902) and Haffkine's dismissal; political control of medicine.
- **Research system** — Kasauli, King Institute, Haffkine Institute, IRFA (1911), *IJMR*, Calcutta School of Tropical Medicine.
- **Ideology** — Curzon: medicine "alone" justifies British rule; but research white-dominated, in hill stations, swamped by vaccine work.

NIGHT-BEFORE REVISION

Quick Revision Sheets

Dates, people, historians and quotes — the "ammunition" that makes an essay look strong. Read these the night before and again on the morning of the exam.

R1. Timeline

YEAR	WHAT HAPPENED	CH
c. AD 400	*Surya Siddhanta* — Indian astronomy text (later dated 11th–13th c. by British critics)	1, 2
1600	English East India Company founded	2
1662	Royal Society of London founded	2
1722–39	Raja Jai Singh's observatories (Jaipur, Delhi, Mathura, Ujjain, Benares)	1, 2
1736	Parsi shipbuilders brought to Bombay; Lowji Nusserwanji Wadia becomes Master Builder	4
1753 / 1757	Linnaeus's *Species plantarum* / Battle of Plassey	2
1767	James Rennell first Surveyor-General	2
1770	Bengal famine (~10 million deaths)	2
1778	J. G. Koenig first naturalist employed by the Company; *Britannia* built at Bombay	2, 4
1784	Asiatic Society of Bengal founded (Sir William Jones)	2
1786–87	Kyd proposes Calcutta Botanic Garden; approved 1787	2
1787	Anton Hove's cotton mission to Gujarat	2, 4
1799	Defeat of Tipu Sultan → Mackenzie's Mysore survey (1799–1810); Lambton begins triangulation	2
1800–14	Francis Buchanan's surveys	2
1818	Great Trigonometrical Survey founded (Lambton); Marathas defeated	2
1820s–31	Bombay shipyard declines after 1820; Calcutta Mint (1831)	4
1829	*Gleanings in Science*; first Indian members of the Asiatic Society	2
1833	Company loses trading rights; Grand Trunk Road rebuilding begins; Porto Novo steel company	2, 4
1834–37	Siwalik fossils (Cautley 1834; Falconer 1837; Wollaston Medal)	2
1835	Macaulay's Minute; Bentinck on promoting European science	2
1835–38	Arthur Cotton's Coleroon anicut	4
1836	Coal Committee; Dwarkanath Tagore buys Raniganj coalfield	2, 4
1839	O'Shaughnessy's telegraph trials	4
1842–54	Ganges Canal (Cautley)	4

1843	Everest completes the Great Arc (Cape Comorin–Dehra Dun, 1,500 miles)	2
1847–52	Godavari anicut (Cotton)	4
1851	Thomas Oldham arrives (Geological Survey of India, 1856); first telegraph lines	2, 4
1853	Dalhousie's railway minute; first railway lines	4
1857–58	Rebellion; Company abolished (1858); Queen's Proclamation	4, 5
1864	Forest Department; Brandis Inspector-General; Aligarh Scientific Society	5
1866	Orissa famine	5
1869	Suez Canal opens; Hunter proposes gazetteers; Sircar's article on cultivation of science	4, 5
1870	Submarine telegraph cable to Britain	4
1871	First all-India census	5
1875	Meteorological Department	5
1876	**Indian Association for the Cultivation of Science** (Mahendralal Sircar)	5
1878	Arms Act; Dehra Dun forestry school; Birdwood's Paris catalogue	4, 5
1880	Famine Commission (20,000 miles of protective railways); Medlicott vs P. N. Bose	4, 5
1881	*Imperial Gazetteer* (9 vols)	5
1890	Botanical Survey of India	5
1893	Voelcker's agriculture report; Haffkine arrives	5
1895–96	J. C. Bose's electric waves; P. C. Ray's mercurous nitrite; Kodaikanal Observatory (1895); plague in Bombay (1896); Tata's science university plan (1896)	5
1897	Venice sanitary conference; Epidemic Diseases Act	5
1898	Elgin admits lack of scientific advisers; Mysore hydro-electric scheme; Muktesar lab	1, 5
1899	Curzon becomes Viceroy (to 1905); Ross resigns; Ray founds Bengal Chemical	5
1901	Famine Commission; Irrigation Commission (1901–03); Havell's handloom proposal	4, 5
1902	Malkowal disaster; Ross's Nobel	5
1903	Board of Scientific Advice; P. N. Bose resigns	5
1905–06	Pusa institute; Swadeshi movement (Partition of Bengal); Kasauli CRI; Indian Agricultural Service	4, 5
1908–09	Indian Institute of Science, Bangalore; Asutosh Mukherjee starts postgraduate science (1909); *Hind Swaraj* (1909)	4, 5
1911–13	IRFA (1911); *Indian Journal of Medical Research* (1913)	5
1914	**Indian Science Congress** founded	5
1916–18	University College of Science (1916); Industrial Commission (Holland)	5
1920–21	P. C. Ray President of Science Congress; Chemical Services Committee; Raman secretary (1921); Calcutta School of Tropical Medicine	5

1930	C. V. Raman wins the Nobel Prize	1, 5
1933	Raman first Indian Director of IISc	5

R2. Who's who — people to name in essays

PERSON	WHY THEY MATTER
Sir William Jones	Founded Asiatic Society (1784); Orientalist; wanted Sanskrit plant names; called "Asiatics" "mere children" in science.
Joseph Banks	President of Royal Society; backed Calcutta garden; saw India as a tropical estate for Britain; predicted tea (1788).
Robert Kyd	Proposed Calcutta Botanic Garden (1786): famine food + profit ("outstrip our rivals").
William Roxburgh	Surgeon-botanist; classified 2,000+ species; costly *Plants of the Coromandel Coast*.
Colin Mackenzie / Francis Buchanan	Great early surveys ("useful or rare or curious"); governmentality.
William Lambton / George Everest	Great Trigonometrical Survey; the Great Arc.
Radhanath Sikdar	Indian worker on the survey (from 1831) — Indian skill but subordinate status.
Hugh Falconer & Proby Cautley	Siwalik fossils; Wollaston Medal 1837. Cautley also built the Ganges Canal.
J. Forbes Royle	Saharanpur garden; tropical/temperate/alpine zones; antiquity of Hindu medicine.
J. D. Hooker	Top British botanist; visited India 1848–50; "chaos of new names"; Darjeeling sanatorium.
Thomas Oldham	Geological Survey of India (1851/56); "tradesman without his tools".
Bal Shastri Jambhekar	Early Indian science populariser in Bombay; Marathi translations; Elphinstone College.
Lowji Nusserwanji Wadia	Parsi Master Builder, Bombay dockyard.
Ardaseer Cursetjee	Parsi engineer; only Indian FRS of the 19th century.
Dwarkanath Tagore	Bought Raniganj coalfield (1836); Agricultural Society member.
W. B. O'Shaughnessy & Shib Chandra Nandy	Telegraph pioneer adapting to tropics & his Indian assistant who built first lines.
Arthur Cotton	Cauvery & Godavari anicuts; credited "old native engineers".
Bentinck / Dalhousie / Curzon	Governors-general who pushed technology: steamers & survey / railways, telegraph ("technophile") / imperial science.
W. W. Hunter	Imperial Gazetteer; "nothing more costly to a Government than ignorance".
Dietrich Brandis	German forester; Inspector-General of Forests (1864); Dehra Dun school.
H. B. Medlicott	Geological Survey head; racist remarks about P. N. Bose and Indians.
Ronald Ross	Malaria–mosquito link; Nobel 1902; frustrated by IMS; resigned 1899.
Waldemar Haffkine	Cholera & plague vaccines; blamed for Malkowal (1902).

Albert & Gabrielle Howard	Pusa; Pusa wheats; respected Indian farming "on its own lines".
Mahendralal Sircar	Founded IACS (1876); science "entirely under native management".
Pramatha Nath Bose	First Indian in Geological Survey; faced racism; found Mayurbhanj iron ore for Tata.
Prafulla Chandra Ray	Mercurous nitrite (1895); Bengal Chemical (1899); "School of Indian Chemistry"; opposed Chemical Service.
Jagadis Chandra Bose	Electric waves (1895–96); plant responses; "the real laboratory is one's own mind".
C. V. Raman	Nobel 1930 (Raman effect); first Indian Director of IISc (1933).
Asutosh Mukherjee	Calcutta VC; postgraduate science (1909); first President of Science Congress (1914).
J. N. Tata	Funded the Indian Institute of Science plan (1896).
Birdwood / Havell	British romantic critics of industrialism; praised crafts & handlooms.
Gandhi / Nehru	*Hind Swaraj*, charkha / "shrine of science".

R3. Historians and their arguments (name-drop these!)

HISTORIAN	ARGUMENT IN ONE LINE	CH
George Basalla (1967)	3-phase diffusion model: data source → dependent colonial science → national science.	1
Michael Adas	Basalla assumed science neutral and its spread inevitable; machines became the "measure of men" (Western superiority).	1, 4
R. K. Kochhar	Basalla "trivialises the compulsions of colonialism".	1
Raina & Habib	Recipient cultures "subvert, contaminate, and reorganise" Western science; *Dawn* debates on science and industry.	1, 5
Roy MacLeod	"Moving metropolis"; Board of Scientific Advice showed a "new sense of scientific independence" (Arnold disagrees).	1, 5
Satpal Sangwan	Colonial scientists were not inferior; Indian shipbuilding declined because of British political pressure.	1, 4
Gyan Prakash	Science became the "organising metaphor" of the Indian elite; science was "translated".	1
Partha Chatterjee	Nationalism challenged colonial rule but accepted its idea of modernity.	1
Ray Desmond	Company's science not purely commercial (Arnold: self-interest dominant).	2
Marika Vicziany	Wellesley used Buchanan's surveys for his reputation; "improvement".	2
C. A. Bayly	The "colonial information order" was built on Indian precursors; Indian agency.	2
Thomas Trautmann	Orientalism fitted India into Western knowledge; Indian astronomy judged by European standards.	2
Matthew Edney	Survey as a "flawed geographic panopticon"; "they did science, the Indians did not"; science marked the European elite.	2

Susan Faye Cannon	The Survey shows British policy better than Macaulay's minute.	2
Andrew Grout	Wollaston Medal was symbolic for Indian geology; Company slowed geology to avoid competition.	2
Daniel Headrick	Technology as "tools of empire"; "British iron" killed Indian shipbuilding.	4
Arnold Pacey	Technology transfer is a "dialogue".	4
Ian Kerr	Railways were built with Indian labour, tools and practices.	4
Ian Stone vs Elizabeth Whitcombe	Canals were beneficial vs canals were "a costly experiment" (waterlogging, salt, malaria).	4
Dharampal	18th-century India had ~10,000 iron furnaces.	4
Deepak Kumar	Racial "apartheid" in science; foresters were "estate-managers".	5
Richard Grove	Company surgeons pioneered conservation; colonies were at the "cutting edge".	5
Gadgil & Guha	Forest policy driven by revenue and railway timber.	5
Michel Foucault (theory)	"Governmentality", "grids" of knowledge, the panopticon.	2, 5

R4. Quotes bank (short and powerful)

QUOTE	WHO / WHERE
Science is "a highly social activity" that cannot be "sealed off from the values of the society in which it is practised".	Opening of Ch 1 (Stepan), p. 1
"The Asiaticks had climbed the heights of science before the Greeks had learned their alphabet."	An Orientalist enthusiast, p. 3
Basalla's model "trivialises the compulsions of colonialism".	Kochhar, p. 14
"Scientific reasoning became the organising metaphor in the discourse of the Western-educated elite."	Gyan Prakash, p. 16
"The British enjoyed the company of science; it would be excessive to suggest that they ruled by it."	Arnold, p. 25
Company science as a "fitful flirtation".	Arnold, p. 25
"A tradesman without his tools or a physician without medicines."	Thomas Oldham, 1856, p. 28
In science, "Asiatics" were "mere children".	Sir William Jones, p. 35
Indian astronomy would "move laughter in girls at an English boarding school".	Macaulay, 1835, p. 38
"By measuring the land … the British distinguished themselves from the Indians: they did science, the Indians did not."	Edney, p. 43
To "outstrip our rivals in every valuable production which nature has confined to this part of the globe".	Robert Kyd, p. 47
"Denial was often as important as diffusion in shaping the technology of colonial India."	Arnold, p. 92

Steam "would be the greatest instrument of civilisation for the people, and of strength for the Government".	Lord Hardinge, 1848, p. 109
Railways: "great engines of social improvement".	Dalhousie, p. 109
"The cheapness and relative abundance of Indian labour was worth many a machine."	Arnold on Ian Kerr, p. 112
"No better employment for Hindostanees than making roads and digging canals."	Begum of Awadh, p. 122
"If the British in India had achieved nothing else, the public works policy … would fully justify their supremacy."	Alfred Deakin, 1893, p. 122
Railways — "beyond dispute that they propagate evil"; machinery "a great sin".	Gandhi, *Hind Swaraj*, 1909, p. 127
"Science existed to serve the empire, not to constitute an alternative source of authority."	Arnold, p. 129
"Nothing [is] more costly to a Government than ignorance."	W. W. Hunter, p. 130
"Secretariat interference … the curse of every Indian branch of special or scientific work."	George Watt, p. 132
Government "almost without competent advisers in many branches of science".	Lord Elgin, 1898, p. 135
Medicine "built on the bed-rock of pure, irrefutable science" — alone a justification for British rule.	Lord Curzon, 1899, p. 136
"He is a Bengali and may be physically unfit for our work."	Medlicott on P. N. Bose, p. 139
"A sort of pestilential fellow with a new bug."	Ernest Hart, 1894, p. 141
"We IMS men are not meant for research … we are simply doctors."	A colleague to Ross, p. 142
Improve Indian agriculture "on its own lines"; Indian practices "worthy of respect".	Albert & Gabrielle Howard, 1910, p. 152
Science has "a higher and nobler claim than the narrow, utilitarian, Benthamite one".	Mahendralal Sircar, p. 159
"The fate of a nation will henceforth depend more upon the achievements of its students of science than upon the skill of its generals."	P. C. Ray, 1920, p. 166
"The real laboratory is one's own mind."	J. C. Bose, p. 167
Bose was "the pioneer who broke this spell".	Birbal Sahni, 1937, p. 168

R5. Numbers that impress examiners

- Asiatic Society members: **119** in 1845 (vs 700+ in London's Geological Society)
- **28 of 42** botanists before 1840 were surgeons
- Great Arc: **1,500 miles** (by 1843)
- Siwalik fossils: **40 tons**; 300 bones in 6 hours
- Bombay ships: **159** (1736–1821)
- Cloth exports: Rs **165** → **8 lakhs** (1816–17 → 1830–31)
- Textile machinery from Britain: **90%+**
- Handlooms: **1/3** of textile output in the 1930s

- Iron furnaces in 18th c.: **~10,000**
- Locomotives: **14,000+** imported vs **~700** made in India
- Railways: **40,000+ miles** by 1947
- Irrigation in 1901: private wells/tanks **~60%**, state canals **~40%**
- Ganges Canal: **1 million acres**
- Scientific services 1920: **195 Europeans vs 18 Indians**
- Pusa 1906: **8 of 40** posts Indian
- Plague deaths: **~8 million** by 1914

IF YOU REMEMBER ONLY ONE PARAGRAPH

Arnold argues that science, technology and medicine in colonial India were **social and political**, not neutral. They were not simply "diffused" from Britain to a passive India; they emerged from an **interaction** with Indian traditions, knowledge, labour and conditions. But this interaction took place within an **unequal colonial relationship**: the British used science for **profit, control and legitimacy** (the "civilising mission"), and **denied** it when it threatened their interests. Under the Company it was a "fitful flirtation"; in the steam age it became the Raj's proud monument; under the Crown it became "imperial science". Indians, meanwhile, made science their own — as a path to **modernity and nationhood** (Sircar, Ray, Bose, Raman), or rejected industrial modernity (Gandhi).